

Using Machine Learning to Compute Constrained Optimal Carbon Tax Rules*

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Abstract

We develop a computational framework for deriving Pareto-improving and constrained Pareto-optimal carbon tax rules in a stochastic overlapping generations (OLG) model with climate change. We assume that carbon taxes are linear in emissions and that the tax rate is a linear function of aggregate state variables. Carbon tax revenue is recycled through a fixed sharing rule. We solve for the optimal coefficients of the tax rule and the optimal carbon revenue-sharing weights. Our approach embeds policy-rule coefficients as pseudo-states in Deep Equilibrium Nets and constructs Gaussian process surrogates, trained via Bayesian active learning, to enable a global search over parameterized tax-and-transfer rules subject to cohort-specific Pareto constraints. We apply the framework to a 12-period OLG model in which both carbon intensity and the climate tipping point are stochastic. An unconstrained welfare-maximizing carbon tax raises social welfare by 1.6% but imposes losses of up to 5% on initial generations. Under Pareto constraints, a simple linear tax on cumulative emissions, paired with optimized intergenerational transfers, increases social welfare by 0.42% in consumption-equivalent terms. Expanding the tax base to include carbon intensity and proximity to climate tipping points raises welfare only marginally, to 0.45%, indicating diminishing returns to policy complexity once intergenerational participation constraints are imposed. The framework offers a scalable template for constrained policy design in heterogeneous-agent models under uncertainty.

Key words: Climate Policy, Optimal Policy, Ramsey Taxation, Pareto Frontier, Deep Learning, Gaussian Processes.

JEL classification: C61, C63, D58, H23, Q54, Q58.

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1 Introduction

Motivation and Challenges. Anthropogenic climate change poses grave risks for current and future generations. As with other externalities, climate change reflects a market failure that can be addressed with appropriate taxes, and carbon taxation is the canonical candidate to correct for this externality. However, it is unclear how to redistribute carbon tax revenues across generations to make all generations better off and thus generate the necessary political support from the current old. Even if one postulates a social welfare function (SWF)¹ that aggregates the welfare of all current and future generations, determining the optimal (SWF maximizing) time-state path for carbon taxes in the presence of heterogeneous agents, stochastic shocks to technology and climate, and nonlinear damage functions remains computationally prohibitive. Most integrated assessment models (IAMs), therefore, either do not consider uncertainty (in which case the so-called social cost of carbon can often be used to determine optimal taxes) or invoke a representative agent, linearize dynamics, or search heuristically over a small set of handpicked tax schedules (see, e.g., Dietz, 2024 and Fernández-Villaverde et al., 2025 for recent reviews). As a consequence, the computational barrier has prevented a systematic search for tax-and-transfer schemes that are not only economically optimal but also politically viable, that is, policies guaranteed to make all generations, including the current, better off.

Objectives and Model Overview. This paper directly confronts this computational barrier. We develop a novel, machine-learning-based framework to systematically search for optimal and Pareto-improving carbon tax schedules for heterogeneous agents exposed to climate risk.² We postulate a standard SWF defined as the weighted sum of expected utilities of all current and future generations and search for carbon taxes and transfers of tax revenue across cohorts that are Pareto-improving and that maximize the SWF. Following most of the literature on optimal carbon taxation, we restrict attention to taxes that are linear in current emissions. The tax rate is nevertheless allowed to vary over time and with economic conditions. While, in principle, a planner could choose an arbitrary state-contingent stochastic process for the tax rate, we instead restrict the tax in each period to be a linear function of key aggregate variables that serve as summary statistics for welfare-improving carbon taxes. In this setting, we examine how (i) constrained-optimal taxes and transfers of tax revenues redistribute risk and welfare across overlapping generations, and (ii) how large the potential welfare gains from *Pareto-improving* policies are. By construction, an allocation that maximizes our SWF is constrained ex ante Pareto-efficient; in that sense, our method identifies a constrained Pareto-efficient carbon tax schedule whose introduction is Pareto-improving relative to the business-as-usual (BAU) allocation (i.e., the no-policy baseline). In this paper, we restrict attention to tax rates that vary linearly with aggregate state variables. While our computational method allows us to easily consider nonlinear functions, it is beyond the scope of this study to carefully examine the welfare effects of higher-order terms. Instead, we focus here on the question of which state variables to include in the sufficient statistic.

Methodological Contribution: A Framework for Constrained Policy Design. The key methodological contribution of this paper is a computational framework that makes constrained welfare analysis feasible in high-dimensional, stochastic overlapping-generations (SOLG) economies. In such environments, policy instruments affect equilibrium allocations across heterogeneous cohorts and over time, and welfare constraints must be satisfied for every cohort across all realizations, not merely at a single point. As a result, identifying tax-and-transfer rules that are both welfare-improving and Pareto-improving faces a fundamental computational barrier, typically requiring either drastic simplifications or ad hoc policy searches.

Our framework overcomes this challenge by recasting the problem of policy design as one of global approximation in the policy space. We build on numerical components developed in earlier work, specifically Deep Equilibrium Nets (DEQNs; see Azinovic et al., 2022) and Gaussian process (GP;

¹For a comprehensive list of acronyms used throughout this paper, please refer to Appendix E.

²Throughout the paper, we assume that agents' welfare is evaluated at an ex ante stage, that is, an agent is identified by the date of their birth and their welfare is evaluated as the time-zero expected utility of their cohort (see, e.g., Gottardi and Kubler, 2011).

see [Rasmussen and Williams, 2005](#)) surrogate models.³ However, the novelty of this paper lies not in a new equilibrium solver but in the application of these tools to constrained policy design.

The central methodological innovation is to treat the coefficients of the tax and transfer rules themselves as parameters that are modeled as pseudo-states within the model solution, alongside the standard economic state variables. By computing equilibrium outcomes globally over the joint space of economic states and policy-rule coefficients, we eliminate the need to recompute the stochastic equilibrium for each candidate policy. This transformation converts the otherwise intractable task of searching over policy rules into a tractable interpolation problem. It provides the computational foundation for imposing and verifying Pareto-improvement constraints during the policy-design stage, using the welfare surrogates introduced below. To the best of our knowledge, this paper is the first in the economics literature to embed parametrized policy rules as pseudo-states in a surrogate framework and to use the resulting surrogates to derive constrained, Pareto-improving policies in a stochastic heterogeneous-agent setting.

The framework proceeds in three integrated steps:

1. **Global solution via policy-conditioned equilibrium surrogates.** We first compute a global solution to the stochastic OLG model across a continuum of tax-and-transfer rules. Let x_t denote the vector of economic and climate state variables relevant for equilibrium determination, and let ϑ denote a finite-dimensional vector of parameters that characterize a given tax-and-transfer rule. We adapt the DEQN solution framework to treat the coefficients of the tax rule $\tau(x_t; \vartheta)$ and the transfer system $\mathbb{T}_{t,j}(x_t; \vartheta)$ as additional inputs alongside the economic state variables. This yields a surrogate representation of equilibrium outcomes that maps any candidate policy vector ϑ directly to equilibrium allocations without requiring the model to be re-solved.
2. **Efficient approximation of welfare objects.** Using the equilibrium surrogate from Step 1, we generate training data and train a GP surrogate model that maps policy parameters ϑ to the SWF and cohort-specific expected utilities. Owing to the data efficiency of GPs, this step produces an accurate and computationally inexpensive approximation of the welfare landscape over the policy space.
3. **Constrained policy optimization.** Finally, we exploit the speed of the welfare surrogate to solve the planner’s non-convex optimization problem. Because both the objective and the Pareto constraints can now be evaluated rapidly, we can systematically search for the tax-and-transfer parameters ϑ^* that maximize the SWF subject to the requirement that no cohort is worse off relative to the business-as-usual allocation.⁴

Taken together, this framework offers a scalable template for policy design in macroeconomic models with uncertainty, heterogeneity, and intergenerational trade-offs, allowing us to identify simple, implementable tax rules that are robust to high-dimensional shocks to carbon intensity and climate damages.

Deep equilibrium networks: intuition. A DEQN provides a flexible global approximation of equilibrium policy functions in high-dimensional dynamic models. Traditional methods often solve models on a fixed grid of states, which becomes computationally impossible as the number of variables grows (the curse of dimensionality). In contrast, DEQNs approximate the equilibrium policy functions (such as consumption or investment rules) directly using a neural network. This network is trained by simulating the economy forward; at each step, if the model’s equilibrium conditions (like Euler equations) are not satisfied, the network adjusts its internal parameters to reduce the error. By

³A “surrogate model”, also known as an emulator or response surface, is a statistical or machine-learning approximation that replicates the input–output behavior of an expensive simulation at a fraction of the computational cost (see, e.g., [Chen et al., 2026](#)).

⁴Once the global equilibrium surrogate is trained, this final optimization step completes in seconds. In contrast, traditional approaches would require re-solving the full stochastic equilibrium for every candidate policy, rendering a systematic search for Pareto-improving rules computationally infeasible (see, e.g., [Brumm et al., 2017](#)).

learning from simulated paths rather than a fixed grid, the algorithm automatically focuses computing power on the economically relevant regions where the economy actually resides. Once trained, the network acts as a fast global surrogate that outputs the equilibrium solution for any given state. For full algorithmic details, see [Azinovic et al. \(2022\)](#).

Gaussian process regression: intuition. While the DEQN allows us to solve the model for a specific tax rule, finding the constrained optimal rule requires searching through thousands of potential parameter combinations. Running a full solution for every combination would be too costly. We therefore use GP regression as a surrogate to approximate the relationship between tax parameters and social welfare. The GP is a flexible probabilistic interpolation technique that “connects the dots” between a small set of expensive model simulations. Crucially, it provides not just a best guess (predictive mean) of welfare, but also a measure of its own uncertainty (predictive variance). This allows us to use Bayesian active learning (BAL): instead of searching blindly, the algorithm intelligently selects the next tax parameters to evaluate based on where the model is most uncertain or most promising ([Renner and Scheidegger, 2018](#)). This drastically reduces the number of expensive simulations required to identify the optimal policy; Appendix C.1 illustrates this mechanism with a simple one-dimensional example (Figure 7). For an implementation of this pipeline in an IAM setting, see [Friedl et al. \(2023\)](#).

Preview of Results. We apply our three-step computational framework to a series of increasingly complex policy design problems, demonstrating its ability to systematically derive and evaluate optimal carbon tax rules. First, we establish a BAU baseline, which, in the absence of policy, projects a mean global warming of approximately 3°C over a 150-year horizon, with worst-case economic damages exceeding 15% of the gross domestic product (GDP).

Our first policy experiment considers a simple, welfare-improving tax rule that is linear in cumulative emissions. The optimal policy under this scheme significantly mitigates climate risk, stabilizing the average global warming at around 2.7°C and substantially reducing the probability of extreme damage scenarios. This policy, however, induces a trade-off across generations: while it increases social welfare by 1.6% in consumption-equivalent terms, the gains are concentrated among future cohorts, whereas the initial generations bear welfare losses of up to 5%.

To address this distributional conflict, in our second experiment, the framework identifies a Pareto-improving policy by simultaneously optimizing the linear tax on cumulative emissions and an accompanying intergenerational transfer scheme, subject to the constraint that all generations must be at least as well off. Our framework successfully identifies a tax-and-transfer rule that ensures no generation is made worse off than under the BAU scenario. Optimal transfers effectively shield the initial cohorts from welfare losses, enabling a positive but modest degree of climate mitigation. This policy yields a social welfare gain of 0.42% in consumption-equivalent terms, without harming any generation, demonstrating that Pareto-improving climate action is achievable within the model. However, constraining policies to make all current and future generations better off necessitates a significantly less ambitious policy path, with much lower taxes, more modest climate mitigation, and substantially smaller welfare gains for future generations.

Finally, we explore whether more complex tax rules provide significant additional benefits. We generalize the tax rule to be a linear function of not only cumulative emissions but also carbon intensity and proximity to a climate tipping point. Although our framework again finds a Pareto-improving policy, the additional welfare gain is minimal. Social welfare rises to 0.45% in consumption-equivalent terms, a marginal improvement of only 7% over the simpler tax-and-transfer scheme. This key result suggests that, in our calibration, the bulk of the welfare gains can be captured by a simple, well-designed tax on cumulative emissions, and that adding further complexity to the tax rule in the sense of adding additional state variables to the summary statistic yields diminishing returns.

Related Literature. This paper is related to three strands of literature: (i) studies analyzing global warming through the lens of OLG models, (ii) research on optimal taxation with heterogeneous agents, and (iii) the growing literature on solving dynamic economic models using machine learning techniques.

(i) Examples of OLG models with environmental concerns include [Burton \(1993\)](#), [Heijdra et al. \(2006\)](#), [Karp and Rezai \(2014\)](#), [Kotlikoff et al. \(2021a\)](#), [Karp et al. \(2024\)](#), and [Kotlikoff et al. \(2024\)](#), all of whom incorporate clean and dirty energy sectors, but abstract from climate uncertainty. Multiple studies address the role of uncertainty in climate policy design. [Golosov et al. \(2014\)](#) show that the optimal carbon tax admits a simple closed-form expression that is largely independent of uncertainty in a representative-agent setting; [Weitzman \(2012\)](#) and [Barnett et al. \(2020\)](#) emphasize the importance of tail risks and ambiguity aversion; and [Kotlikoff et al. \(2021b\)](#) highlight how climate uncertainty interacts with intergenerational equity in OLG models. Our paper is most closely related to [Kotlikoff et al. \(2021b\)](#), who study optimal Pareto-improving taxes in an OLG model with climate uncertainty. While we abstract from modeling clean and dirty energy sectors separately, we expand on their work by proposing a global solution method to compute constrained Pareto-optimal, state-dependent carbon taxes in an OLG model with climate uncertainty.

(ii) We contribute to the literature on optimal Ramsey problems in quantitative, heterogeneous-agent models. [Dyrda and Pedroni \(2023\)](#), [Bhandari et al. \(2013\)](#), and [Feng et al. \(2024\)](#) analyze optimal fiscal policy in incomplete-markets models with heterogeneous agents. [Itskhoki and Moll \(2019\)](#) characterize the Ramsey optimal development policy. [Nuño and Thomas \(2020\)](#) derive the Ramsey optimal monetary policy. [Douenne et al. \(2024\)](#) are closely related to our work in that they study optimal climate policy in a dynamic general equilibrium model with heterogeneous agents and linear tax instruments. The key differences lie in both the economic environment and the computational approach. [Douenne et al. \(2024\)](#) analyze an Aiyagari-style infinite-horizon economy and characterize constrained optimal policy for a given social welfare function, whereas we consider a finite-horizon overlapping-generations framework with aggregate uncertainty and focus on identifying Pareto-improving tax-and-transfer rules across cohorts. This setting gives rise to a policy problem with cohort-specific participation constraints. Our main contribution is to introduce a scalable computational framework for constrained policy design into this literature.

(iii) We also contribute to the rapidly growing literature that employs machine learning techniques to solve dynamic stochastic models and to address optimal policy problems.⁵ We combine methods developed in [Scheidegger and Bilonis \(2019\)](#), [Azinovic et al. \(2022\)](#), and [Chen et al. \(2026\)](#). Conceptually, our workflow resembles that of [Friedl et al. \(2023\)](#): they (a) use DEQNs to solve a stochastic planner’s problem and (b) fit a Gaussian process surrogate to the social cost of carbon, which they then use for parametric uncertainty quantification ([Harenberg et al., 2019](#)). Our framework departs from theirs in three key ways. First, we use a SOLG model rather than a representative-agent setting. Second, rather than treating structural parameters such as risk aversion as pseudo-states, we embed the coefficients of the carbon-tax and transfer rules directly in the state vector and solve for the resulting policy functions with a single DEQN run. Third, we construct GP surrogates for the SWF, not for the social cost of carbon, and embed this surrogate in an optimizer to recover welfare-optimal taxes and transfer schedules.

Organization of the Article. Section 2 introduces a SOLG framework with an explicit climate externality. Section 3 presents the calibration. Section 4 develops a generic, machine-learning-based procedure for computing Pareto-improving policies in heterogeneous-agent economies with various sources of uncertainty. Section 5 applies the method to a series of numerical experiments, and Section 6 concludes.

2 A Stochastic OLG Model with Climate Change

We consider a SOLG economy with climate change caused by industrial emissions that reduces total factor productivity through a damage function. There is an infinite time-horizon, $t = 0, \dots$, and exogenous Markovian shocks are realized every period. The economic block consists of 12 overlapping generations of selfish agents, in which each period spans five calendar years; households enter the

⁵See [Fernández-Villaverde et al. \(2024\)](#) for a recent review of neural-network-based algorithms in macroeconomics, and [Garnett \(2023\)](#) for approaches that use GPs for optimization.

economy at age 20 and exit at age 80 (Kotlikoff et al., 2021b). A representative firm produces a single consumption good using capital and labor as inputs and emits CO₂ during production. We treat carbon intensity as an exogenous stochastic process that declines over time in expectation. The climate block is represented by a reduced-form, cumulative-emissions emulator (Brock and Xepapadeas, 2017; Dietz and Venmans, 2019).⁶ Climate damages follow the convex specification of Weitzman (2012), augmented with stochastic tipping points in the spirit of Kotlikoff et al. (2021b).

The only sources of uncertainty in our model are the stochastic process governing carbon intensity and the stochastic nature of the tipping point. This specification abstracts from several other quantitatively important dimensions of climate uncertainty, most notably uncertainty about climate sensitivity and the damage function. These sources of uncertainty are more appropriately modeled within a framework that incorporates learning, which would go beyond the scope of this study (see, e.g., Kelly and Kolstad, 1999; Friedl et al., 2023, and references therein).

We outline the model in the following steps: Section 2.1 presents the firm block, Section 2.2 explains household behavior, Section 2.3 details the role of the government, Section 2.4 describes the climate module, and Section 2.5 defines the competitive equilibrium. The calibration of all model parameters is presented in Section 3.

2.1 Firms

The representative firm produces the single consumption good Y and emissions e_t at each time t :

$$(Y_t, e_t) = f_t(K_t, L_t, \mu_t) = (\Omega_t(T_t^{AT})\Phi(1 - \mu_t)K_t^\alpha L_t^{1-\alpha} + (1 - \delta)K_t, (1 - \mu_t)\kappa_t K_t^\alpha L_t^{1-\alpha}). \quad (1)$$

Using the emissions definition, we can equivalently express abatement in terms of emissions. In particular, emissions satisfy

$$e_t = (1 - \mu_t) \kappa_t K_t^\alpha L_t^{1-\alpha}, \quad (2)$$

which can be inverted to obtain

$$1 - \mu_t = \frac{e_t}{\kappa_t K_t^\alpha L_t^{1-\alpha}}. \quad (3)$$

Substituting (3) into the production function yields an equivalent representation of output as

$$Y_t = \Omega_t(T_t^{AT}) \Phi\left(\frac{e_t}{\kappa_t K_t^\alpha L_t^{1-\alpha}}\right) K_t^\alpha L_t^{1-\alpha} + (1 - \delta)K_t. \quad (4)$$

This formulation makes explicit that, conditional on factor inputs, output depends on emissions through the implied abatement rate. Furthermore, it demonstrates that e_t can also be interpreted as an energy input of the production function. Feasibility follows from the constraint $0 \leq \mu_t \leq 1$, which ensures that the argument of $\Phi(\cdot)$ lies in $[0, 1]$.

Production is given by a Cobb-Douglas production function, where α is the output elasticity of capital, K , L , and δ are capital, labor, and depreciation, respectively. μ denotes abatement, and $\Phi : [0, 1] \rightarrow [0, 1]$ is a mitigation-cost function. κ is the exogenous carbon intensity of production and $\Omega(T^{AT})$ is the damage function arising from climate change, where T^{AT} denotes the global mean temperature anomaly above pre-industrial levels. Following Nordhaus (2017), we assume an abatement function of the form

$$\Phi(1 - \mu) = 1 - \theta_1 \mu^{\theta_2}. \quad (5)$$

Note that when $e_t = 0$ (equivalently $\mu_t = 1$), output is reduced by a factor of θ_1 .

Firms take as given a carbon tax rate τ_t and maximize profits:

$$\max_{\mu_t, K_t, L_t} \Pi = \Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) K_t^\alpha L_t^{1-\alpha} + (1 - \delta)K_t - w_t L_t - (1 + r_t)K_t - \tau_t(1 - \mu_t)\kappa_t K_t^\alpha L_t^{1-\alpha} \quad (6)$$

⁶Climate emulators are simplified models that link CO₂ emissions to global mean temperature at negligible computational cost and are therefore well suited for policy analysis. For more details, see, e.g., Dietz et al. (2021); Folini et al. (2024), and Eftekhari et al. (2025).

$$\text{s.t. } 0 \leq \mu_t \leq 1.$$

The carbon intensity (κ_t) follows an exogenous time-varying stochastic process. In the calibration, our objective will be to capture the uncertainty in CO₂ emissions and account for the possibility that emissions remain elevated over the next 100 years. Unabated emissions lead to an increase in atmospheric temperature, which triggers damages that reduce output. We employ a damage function as in [Weitzman \(2012\)](#) and follow [Kotlikoff et al. \(2021b\)](#) by augmenting damages with stochastic tipping:

$$\Omega_t(T_t^{AT}) = \frac{1}{1 + \left(\frac{1}{\psi_1} T_t^{AT}\right)^2 + \left(\frac{1}{2 \cdot TP_t} T_t^{AT}\right)^{6.754}}, \quad (7)$$

where T_t^{AT} is the atmospheric temperature anomaly and TP_t is the tipping point, which follows a random walk.⁷ While climate tipping damages are negligible below the threshold, [Weitzman \(2012\)](#) notes that once the tipping point is reached, an exponent of 6.754 causes these damages to dominate the damage function.

2.2 Households

In each period, a new household enters the economy. Households live for $A = 12$ periods, where each period corresponds to 5 years. There is no uncertainty regarding lifetime. An agent born at time t maximizes expected lifetime utility given by:

$$\mathbb{E}_t \sum_{j=1}^A \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}, \quad (8)$$

subject to:

$$C_{t,j} + a_{t+1,j+1} = (1 + r_t)a_{t,j} + w_t l_j + \mathbb{T}_{t,j}, \quad (9)$$

where β is the discount factor, $C_{t,j}$, $a_{t,j}$, l_j are consumption, assets, and labor endowment of an agent of age j at time t . r_t and w_t are the factor prices of production (interest rate and wages), and $\mathbb{T}_{t,j}$ is the transfer from the government.

In each period, households sell their assets to the firm and supply labor inelastically. Households choose consumption $C_{t,j}$ and savings $a_{t+1,j+1}$ in period t . We follow [Benzell et al. \(2015\)](#) and assume that agents enter the workforce with 0 assets, work for 8 periods (40 years), and then retire. After retirement, agents' labor endowment drops to 40% of the endowment in period 8, which serves as a proxy for a pension system. Further details on the recursive formulation of the household problem are given in [Appendix A.1](#).

We define the time-zero expected utility of a generation born at time t as

$$\tilde{U}_t = \mathbb{E}_0 \sum_{j=1}^A \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}, \quad (10)$$

where $C_{t+j-1,j}$ denotes the agent's optimal consumption under a given policy scenario. $\tilde{U}_t$ distinguishes welfare under a given policy from the utility in the BAU case, denoted by U_t , which assumes zero taxes and is defined analogously.

⁷A tipping point is a critical threshold in the climate system beyond which a small additional change triggers large, abrupt, and often irreversible shifts in the state of the system (examples include a collapse of the Atlantic Meridional Overturning Circulation, Amazon rainforest dieback, or disintegration of the West Antarctic Ice Sheet; see, e.g., [Lenton et al., 2008](#); [Cai and Lontzek, 2018](#)). Note that, whereas we refer to a tipping point in [Equation \(7\)](#), [Dietz and Asheim \(2012\)](#) consider such a damage specification, with a changing exponent, as an instance of epistemic uncertainty.

2.3 Government

The government levies a carbon tax on firms and redistributes the resulting tax revenue among contemporaneous households.

General tax-and-transfer rule. We study parsimonious and interpretable policy rules in a stochastic OLG economy. To keep the policy space tractable and transparent, we restrict attention to a class of parametric tax-and-transfer rules that map the aggregate economic–climate state vector x_t to a carbon tax rate and a contemporaneous transfer scheme. Specifically, policies take the form

$$\tau_t = \tau(x_t; \vartheta), \quad \mathbb{T}_{t,j} = \mathbb{T}_{t,j}(x_t; \vartheta), \quad (11)$$

where ϑ is a finite-dimensional vector of policy parameters. Transfers are specified at the cohort (age-group) level and should be interpreted as stylized age-dependent redistribution schemes (for example, via taxes, rebates, or pension adjustments). We abstract from within-cohort heterogeneity, administrative costs, and political frictions and adopt a narrow, welfare-based notion of implementability. The government budget is balanced period by period:

$$\sum_{j=1}^A \mathbb{T}_{t,j} = \tau_t e_t \quad \forall t. \quad (12)$$

All policy experiments studied in Section 5 are special cases of (11), obtained by restricting the arguments of $\tau(\cdot)$ and $\mathbb{T}_{t,j}(\cdot)$ and the dimension of ϑ . Throughout the paper, we treat the economic and climate calibration as given and do not propagate parametric uncertainty through the policy design problem; a complementary analysis focusing on parametric uncertainty quantification in IAMs, rather than constrained policy design, is provided in Friedl et al. (2023).

Planner’s problem. The government chooses the policy parameters ϑ in period 0 to maximize ex-ante social welfare. Let $\tilde{U}_t(\vartheta)$ denote the expected lifetime utility, evaluated at time 0, of the cohort born in period t under the policy parametrized by ϑ . The planner solves

$$\max_{\vartheta} \sum_{t=-10}^{29} \gamma_t \tilde{U}_t(\vartheta), \quad (13)$$

where γ_t are welfare weights and the summation ranges over all cohorts alive at $t = 0$ as well as those born over the subsequent 150 years.

We evaluate welfare changes using the consumption-equivalent variation

$$\left(\frac{\tilde{U}_t(\vartheta)}{U_t} \right)^{\frac{1}{1-\sigma}} - 1, \quad (14)$$

where U_t denotes the expected lifetime utility of cohort t under the BAU allocation. For constant relative risk aversion (CRRA) preferences with $\sigma > 1$, this transformation yields a well-defined and monotone measure of welfare changes across policies.

In addition to the unconstrained problem (13), we consider a Pareto-improvement constraint requiring that every cohort is weakly better off relative to BAU:

$$\left(\frac{\tilde{U}_t(\vartheta)}{U_t} \right)^{\frac{1}{1-\sigma}} - 1 \geq 0, \quad \forall t = -10, \dots, 29. \quad (15)$$

2.4 Climate Module

Climate science has established a near-linear relationship between cumulative CO₂ emissions and the resulting global warming. We adopt this relationship:

$$T_t^{AT} \approx \sigma_{CCR} \sum_{s=0}^{t-1} e_s, \quad (16)$$

where T_t^{AT} denotes the global mean surface temperature anomaly above pre-industrial levels, e_s represents global emissions, and σ_{CCR} is the constant of proportionality, referred to as the carbon-climate response (CCR) or the Transient Climate Response to Cumulative Carbon Emissions (TCRE; [Matthews et al. \(2009\)](#)). Within economics, the value of this simplifying approximation was recognized early on by [Brock and Xepapadeas \(2017\)](#) and [Dietz and Venmans \(2019\)](#). We report emissions in terms of CO₂. For notational convenience, we also define the stock of carbon in the atmosphere as

$$E_t = \frac{1}{3.666} \sum_{s=0}^{t-1} e_s, \quad (17)$$

where 3.666 is the conversion factor from carbon (C) to CO₂. Additional details can be found in [Appendix A.2](#).⁸

2.5 Equilibrium

A competitive equilibrium, given initial conditions $(E_0, T_0^{AT}, TP_0, \kappa_0, \{a_{0,j}\}_1^A)$, a tax function (e.g., $\tau(E, \kappa, TP)$) and transfer scheme ($\mathbb{T}$), is a collection of choices for households $(\{C_{t,j}\}_{j=1}^A, \{a_{t+1,j+1}\}_{j=1}^A)_{t=0}^\infty$, the representative firm $(K_t, L_t, \mu_t)_{t=0}^\infty$, and prices $(r_t, w_t)_{t=0}^\infty$ such that:

1. Given $(r_t, w_t, \mathbb{T})_{t=0}^\infty$, households maximize [\(8\)](#) subject to [\(9\)](#).
2. Given $(r_t, w_t, \tau_t)_{t=0}^\infty$, firms maximize [\(6\)](#).
3. All markets clear, that is,

$$\sum_{j=1}^A a_{t,j} = K_t \quad \forall t, \quad (18)$$

$$\sum_{j=1}^A l_j = L_t \quad \forall t. \quad (19)$$

The first-order conditions of the firm imply that

$$\Omega_t(T_t^{AT})(\theta_1 \theta_2 \mu_t^{\theta_2-1}) = \tau_t \kappa_t, \quad (20)$$

$$r_t = \alpha K_t^{\alpha-1} L_t^{1-\alpha} \left(\Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t(1 - \mu_t) \right) - \delta, \quad (21)$$

$$w_t = (1 - \alpha) K_t^\alpha L_t^{-\alpha} \left(\Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t(1 - \mu_t) \right). \quad (22)$$

The Euler equations for households $j \in \{1, \dots, A-1\}$ are given by:

$$C_{t,j}^{-\sigma} = \beta \mathbb{E}_t \left[(1 + r_{t+1}) C_{t+1,j+1}^{-\sigma} \right]. \quad (23)$$

Note that there is no Euler equation for $j = A$ because agents consume their entire wealth in the final period.

⁸We report cumulative emissions in thousands of gigatonnes of carbon (GtC) and, for brevity, use the terms “stock of carbon” and “cumulative emissions” interchangeably.

3 Calibration

This section presents the model calibration, and the corresponding parameter values are summarized in Table 1.⁹

Households. Households live for 12 periods, where each period corresponds to 5 years. Annual labor endowments follow the functional form proposed by Benzell et al. (2015):

$$l_i = \begin{cases} l_0 e^{4.47+0.033i-0.00067i^2} & i \leq 40, \\ l_i = 0.4 \cdot l_{40} & i > 40, \end{cases} \quad (24)$$

where i denotes the age in years of a given cohort (distinct from the five-year periods denoted by j). The constant l_0 is chosen such that the aggregate labor endowment is normalized to a value of 1. The preference parameters are standard and can be found in Table 1. To obtain the initial asset distribution, we simulate a deterministic model with constant temperature and no climate change, using its equilibrium as the initial distribution.

Production. The production function is Cobb-Douglas with standard parameters (see Table 1). We consider uncertainty in the carbon intensity of production κ_t . We model carbon-intensity uncertainty as a non-negative autoregressive process:

$$\kappa_t = \begin{cases} \rho_t \kappa_{t-1} + \epsilon_{\kappa,t}, & \text{if } \rho_t \kappa_{t-1} + \epsilon_{\kappa,t} > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (25)$$

where the innovation $\epsilon_{\kappa,t}$ is i.i.d. and takes the following discrete values:

$$\epsilon_{\kappa,t} = \begin{cases} 0.03, & \text{with probability } \frac{1}{3}, \\ 0.00, & \text{with probability } \frac{1}{3}, \\ -0.03, & \text{with probability } \frac{1}{3}. \end{cases} \quad (26)$$

Thus, carbon intensity κ_t follows a bounded random walk that cannot fall below zero: once it reaches zero, it remains at that absorbing state.

The persistence parameter ρ_t evolves deterministically according to

$$\rho_t = \rho_0 + (\rho_\infty - \rho_0) \left(1 - e^{-\delta^\rho \Delta_t t}\right), \quad (27)$$

where ρ_0 denotes the initial level, ρ_∞ its long-run limit, $\Delta_t = 5$ is the length of one period in years, and δ^ρ governs the speed of convergence to that limit (see Table 1 for their numerical values).

Following Nordhaus (2017), we initialize carbon intensity at $\kappa_0 = 0.35032$ and set the initial stock of cumulative emissions at 851 GtC. Total factor productivity is calibrated to $600^{1-\alpha}$; this scaling factor matches base-year CO₂ emissions to observed data. Figure 2 below shows the resulting BAU emission path together with its uncertainty band. Note that average emissions closely match the RCP 4.5 emissions scenario. Specifying uncertainty in carbon intensity is one of the most challenging issues in climate-change macroeconomics (see Barnett et al., 2022 for an in-depth analysis).

We set the mitigation-cost parameters to $\theta_1 = 0.7$ and $\theta_2 = 2.6$. While θ_2 follows the estimate in Nordhaus (2017), the value of θ_1 may appear high. However, as discussed in Section 2.1, θ_1 governs the output loss associated with full abatement, that is, when industrial carbon emissions are reduced to zero. In this context, a relatively high value of θ_1 is plausible and ensures that full abatement is never optimal in the model (see also the discussion emphasizing the dominance of climate damages over direct emissions-related costs in Barnett et al., 2022).

⁹While systematic uncertainty quantification is beyond the scope of this study, we examine the sensitivity of our main results to alternative values of the abatement-cost parameter θ_1 and the carbon-intensity process κ_t in Appendix D.

	Symbol	Parameter	Value
Households	β	Discount factor	0.9
	σ	Relative risk aversion	3
	L	Aggregate labor (to match initial emissions)	600
Firms	α	Capital share	0.3
	δ	Depreciation	0.2
	κ_0	Initial carbon intensity	0.35032
	ρ_0	Initial AR(1) param for carbon intensity	1.08
	ρ_∞	Final AR(1) param for carbon intensity	0.91
	δ^ρ	AR(1) param depreciation for carbon intensity	0.04
	θ_1	Intercept of abatement cost function	0.7
	θ_2	Exponent of abatement cost function	2.6
Climate	E_0	Initial stock of carbon (thousands of GtC)	0.851
	σ_{CCR}	Climate sensitivity ($^\circ\text{C}$ per thousand GtC)	1.7
	ψ_1	Parameter damage function	13.16

Table 1: Summary of Model Parameters.

Climate Module. To obtain roughly 3°C of average warming under BAU while approximating the Representative Concentration Pathway (RCP) 4.5 emissions path, we calibrate the carbon-intensity parameters to $\rho_0 = 1.08$, $\delta^\rho = 0.04$, and $\rho_\infty = 0.91$.

Damages. Following [Kotlikoff et al. \(2021b\)](#), we model climate damages with a stochastic tipping point. The threshold TP_t evolves as a bounded random walk provided that the contemporaneous atmospheric temperature T_t^{AT} remains below the threshold,

$$TP_t = TP_{t-1} + \epsilon_{TP,t}, \quad (28)$$

where the innovation $\epsilon_{TP,t}$ is i.i.d. and takes the following discrete values:

$$\epsilon_{TP,t} = \begin{cases} 0.1, & \text{with probability } \frac{1}{3}, \\ 0, & \text{with probability } \frac{1}{3}, \\ -0.1, & \text{with probability } \frac{1}{3}. \end{cases} \quad (29)$$

Reflective barriers at $\underline{TP} = 2.5$ and $\overline{TP} = 3.5$ prevent implausible extremes. Once the realized temperature reaches the threshold ($T_t^{\text{AT}} \geq TP_t$), the tipping point becomes absorbing: $TP_{t+s} = TP_t$ for all $s \geq 0$. The curvature parameter of the damage function is set to $\psi_1 = 13.16$.

4 Machine Learning for Computing Constrained Pareto Optimal Policies

Methodological positioning. The computational components used in this section build on existing methods for solving and emulating high-dimensional dynamic models. The novelty of the present framework does not lie in a new equilibrium solver, but in the treatment of policy-rule coefficients as pseudo-states, which allows us to compute equilibrium allocations and welfare outcomes globally over a space of implementable tax-and-transfer rules and thereby render constrained Pareto-improving policy design computationally tractable. Building on this idea, the section develops a generic, three-step machine-learning-based methodology for computing welfare-maximizing or constrained Pareto-optimal policies in stochastic models with heterogeneous agents. Figure 1 provides a schematic overview of the three-step procedure, which we describe in detail in Sections 4.1–4.2.3.

Step 1: Deep Equilibrium Nets Surrogate. In Step 1 (cf. Section 4.1 for details), we employ DEQNs (Azinovic et al., 2022) to solve the SOLG IAM over a continuum of parameterized tax (and transfer) schemes by treating their parameters as pseudo-states in the model’s state vector (Chen et al., 2026). This formulation enables the simultaneous computation of global solutions for all feasible schemes in a single model run, yielding policy functions as explicit mappings:

$$P: (x_{\text{state},t}, \vartheta) \mapsto \mathbb{R}^M, \quad (30)$$

where $x_{\text{state},t} \in \mathbb{R}^{d_x}$ is the economic-climate state vector, and $\vartheta \in [a, b] \subset \mathbb{R}^{d_\vartheta}$ are the policy parameters. We define the augmented state:

$$x_t = \begin{bmatrix} x_{\text{state},t} \\ \vartheta \end{bmatrix} \in \mathbb{R}^{d_x + d_\vartheta}, \quad (31)$$

with overall problem dimension $d = d_x + d_\vartheta$. In our specific implementation of the SOLG model (see Section 5), the augmented state vector reads as:

$$x_t = [t, TP_t, TP_{\text{reached},t}, \kappa_t, \{a_{t,j}\}_{j=1}^A, E_t, \vartheta]^\top \in \mathbb{R}^{A+5+d_\vartheta}, \quad (32)$$

where we define $TP_{\text{reached},t}$ as an absorbing indicator that equals one once the tipping point has been reached and remains zero otherwise. The block ϑ collects the pseudo-state variables, whose dimensions and interpretations depend on the tax and transfer scheme being analyzed. For example, in Section 5.2 we study a linear cumulative-emissions tax with parameters $\vartheta = (\vartheta_0, \vartheta_E) \in \mathbb{R}^2$, for which

$$\tau_t = \vartheta_0 + \vartheta_E E_t. \quad (33)$$

While the preceding example is illustrative, we solve much richer model specifications below. The SOLG model discussed in Section 5.4, for instance, assumes a tax policy based on cumulative emissions, carbon intensity, climate tipping, and optimal transfers, thereby introducing 16 additional pseudo-state variables, resulting in a state space of dimension $d = 33$. We approximate the policy by a neural network, that is, $P(x_t) \approx \mathcal{N}(x_t)$, trained to generate the $2(A - 1) = 22$ -dimensional output of the SOLG (cf. Appendix A.3):

$$\mathcal{N}(x_t) = [\{a_{t+1,j+1}\}_{j=1}^{A-1}, \{v_{t,j}\}_{j=1}^{A-1}]^\top \in \mathbb{R}^{22}, \quad (34)$$

which concatenates the next-period asset choices and contemporaneous, normalized value functions for all cohorts $j = 1, \dots, A - 1$ (see Appendices A.1 and A.3 for more details). Training the DEQN is computationally demanding; it typically requires several hours on a standard laptop, but once fitted, policy evaluations (i.e., interpolating the trained neural network for a single state) and forward simulations (i.e., chaining these queries to generate a full time path) are virtually cost-free, completing in a few seconds. This illustrates the key advantage of surrogate models: the computational burden is front-loaded, and the one-off (seemingly high) cost is quickly amortized over subsequent analyses.¹⁰

Step 2: Gaussian-process Social Welfare Function Surrogate. In Step 2 (see Section 4.2 for details), we fit a Gaussian Process Regression (GPR) surrogate that approximates the expected SWF,

¹⁰For perspective, state-of-the-art solvers for stochastic OLG models, which do not include a climate block, such as the time-iteration method with adaptive sparse grids (see Scheidegger et al., 2018, and references therein) require on the order of tens of thousands of core-hours to solve a single calibration (i.e., a fixed set of parameters) that features roughly the same number of state variables and policy functions as the largest model we consider (cf. Section 5.4). In our framework, Step 3 evaluates thousands of tax–transfer parameter vectors; consequently, employing conventional techniques would render a systematic search for welfare- and Pareto-improving tax–transfer schemes computationally infeasible.

our quantity of interest (QoI), $\widehat{\text{QoI}}$, as a function of the policy parameter vector ϑ :

$$\widehat{\text{QoI}}: \vartheta \mapsto \mathbb{R}. \quad (35)$$

For example, in the welfare-maximization exercise studied in Section 5.2, we consider $\widehat{\text{QoI}}(\vartheta) := \mathbb{E}[W(\vartheta)]$, the expected value of the social-welfare function under the tax scheme parametrized with ϑ . The training data for these GPR surrogates are generated through computationally relatively expensive simulations that leverage the policy functions $P(x_t) \approx \mathcal{N}(x_t)$, computed in Step 1 using DEQNs. To mitigate the relatively high computational cost of generating training data, we employ GPR due to its ability to accurately interpolate the SWF using only a limited number of efficiently selected simulation points via BAL, a form of reinforcement learning (see Renner and Scheidegger, 2018, and references therein).

Step 3: Optimization of Welfare Parameters. In Step 3 (see Section 4.2 for details), the GPR surrogate’s low evaluation cost enables rapid solution of the planner’s optimization problem:

$$\begin{aligned} \vartheta^* &= \arg \max_{\vartheta \in [a,b]} \widehat{\text{QoI}}(\vartheta), \\ &\text{s.t. constraints.} \end{aligned} \quad (36)$$

Given the constraints and the GP surrogate objective obtained in Step 2, we can identify welfare-improving (Pareto-optimal) policy parameters ϑ^* for a given tax-and-transfer scheme at virtually no additional computational cost: the optimization is performed directly on the GP surrogate, without the need for further model simulations, and thus completes in seconds.¹¹

Figure 1 provides a schematic overview of the three-step procedure outlined above. Sections 4.1–

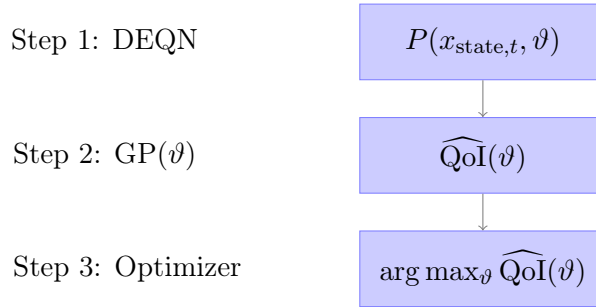


Figure 1: Schematic of the solution scheme: first, DEQNs are used to solve the SOLG to obtain the optimal policy, $P(x_{\text{state},t}, \vartheta) \approx \mathcal{N}(x_{\text{state},t}, \vartheta)$, as a function of the economic and climate state variables, as well as the parameters (which have been added to the state space as so-called pseudo-state variables); second, a GP surrogate is trained on the SWF, simulated from $\mathcal{N}(x_{\text{state},t}, \vartheta)$; third, an optimizer finds the policy parameters ϑ^* that maximize $\widehat{\text{QoI}}(\vartheta)$.

4.2.3 provide the implementation details underlying the three blocks in Figure 1; for the reader’s convenience, we summarize the full operational checklist in Box 4.2.3 at the end of this section.

4.1 Deep Equilibrium Nets

In this section, we provide a concise summary of the general concept of DEQNs, following the notation used by Azinovic et al. (2022) and Nuño et al. (2024). DEQNs are a simulation-based approach that employs deep neural networks to approximate the optimal policy function $P: X \rightarrow Y \subset \mathbb{R}^M$ for a dynamic model. This method assumes that the underlying economy can be described by discrete-time

¹¹ Appendix B provides further details on the model-dependent constraints for Equation (36) and the permissible ranges for the pseudo-states ϑ .

first-order equilibrium conditions, expressed as:

$$\mathbf{G}(x_t, P) = 0, \forall x_t \in X \subset \mathbb{R}^d. \quad (37)$$

Neural-network approximation. DEQNs approximate the unknown policy function using a neural network. Specifically, the policy function $P(x)$ is approximated as $P(x_t) \approx \mathcal{N}(x_t)$, where $\mathcal{N}(\cdot)$ is a neural network with trainable parameters ν . These parameters are initially unknown and are determined by optimizing a suitable loss function that evaluates the accuracy of the approximation at a given economic state.

In our SOLG application, we use densely connected feed-forward neural networks (FNN). In accordance with the existing literature, we define an L -layer FNN as a function $\mathcal{N}^L(x_t) : \mathbb{R}^{d_{\text{input}}} \rightarrow \mathbb{R}^{d_{\text{output}}}$, where the network consists of $L-1$ hidden layers. The ℓ -th layer contains N_ℓ neurons, with $N_0 = d_{\text{input}}$ and $N_L = d_{\text{output}}$. For each layer $1 \leq \ell \leq L$, we introduce a weight matrix $\mathbf{W}^\ell \in \mathbb{R}^{N_\ell \times N_{\ell-1}}$ and a bias vector $b^\ell \in \mathbb{R}^{N_\ell}$. The affine transformation in the ℓ -th layer is defined as $A^\ell(x_t) = \mathbf{W}^\ell x_t + b^\ell$. Using a nonlinear activation function $\sigma(\cdot)$, such as ReLU, Swish, or GELU, the FNN can be expressed as:

$$P(x_t) \approx \mathcal{N}(x_t) = \mathcal{N}^L(x_t) = A^L \circ \sigma_{L-1} \circ A^{L-1} \circ \dots \circ \sigma_1 \circ A^1(x_t). \quad (38)$$

The selection of hyperparameters $\{L, \{N_\ell\}_{\ell=1}^L, \{\sigma_\ell(\cdot)\}_{\ell=1}^L\}$ is known as the architecture design. Approaches to tuning these hyperparameters range from prior-experience heuristics and manual, random, or grid search to more sophisticated techniques such as Bayesian optimization (see [Elsken et al. \(2019\)](#) for a comprehensive review).

Training via Simulated Paths. In the DEQN algorithm, an initial guess, by randomly initializing the FNN, is made for the unknown policy function. Following this, a sequence of $N_{\text{path length}}$ states is simulated. Starting from a given state x_t , the subsequent state x_{t+1} is determined by the policies encoded by the neural network, $\mathcal{N}(x_t)$, and the remaining model-implied dynamics.

If the approximate policy function satisfying the equilibrium conditions were known, Equation (37) would hold along a simulated path. However, since the neural network is initialized with random coefficients, $\mathbf{G}(x_t, \mathcal{N}(x_t)) \neq 0$ along the simulated path of length $N_{\text{path length}}$. This residual is used to refine the approximate policy function.

Specifically, DEQNs employ a loss function that quantifies the error in the equilibrium conditions, defined as:

$$\ell_\nu = \frac{1}{N_{\text{path length}}} \sum_{x_t \text{ on sim. path}} \sum_{m=1}^{N_{\text{eq}}} (\mathbf{G}_m(x_t, \mathcal{N}(x_t)))^2, \quad (39)$$

where $\mathbf{G}_m(x_t, \mathcal{N}(x_t))$ represents all the N_{eq} first-order conditions of a given model (as outlined in Section 2), expressed as

$$\mathbf{G}(x_t, \mathcal{N}(x_t)) = \sum_{m=1}^{N_{\text{eq}}} \mathbf{G}_m(x_t, \mathcal{N}(x_t)). \quad (40)$$

Equation (39) is utilized to update the weights of the network using any variant of (stochastic) gradient descent:

$$\nu'_k = \nu_k - \alpha \frac{\partial \ell(\nu)}{\partial \nu_k}, \quad (41)$$

where ν_k denotes the network weights, ν'_k the updated weights, α is the learning rate, and $\frac{\partial \ell(\nu)}{\partial \nu_k}$ is the gradient of the loss function with respect to the weights.¹² The updated neural network-based approximation of the policy functions is subsequently used to simulate the model again and evaluate the loss function. This iterative procedure is pursued until $\ell_\nu < \epsilon$ for some tolerance $\epsilon > 0$, at which point an approximate equilibrium policy has been found. In our numerical experiments (see Section 5),

¹²In our numerical experiments, we employ the Adam optimizer ([Kingma and Ba, 2014](#)), a variant of stochastic gradient descent.

we set $\epsilon = 10^{-6}$.

Four Key Ingredients. To summarize, the DEQN algorithm comprises four components: (i) a neural network approximator such as an FNN (cf. Equation (38)), (ii) the equilibrium-error loss (cf. Equation (39)), (iii) a gradient-based update rule (cf. Equation (41)), and (iv) simulated paths that supply training states.

SOLG-specific Modifications to DEQN. To adapt the baseline DEQN framework (Azinovic et al., 2022) for solving non-stationary SOLG IAMs across a continuum of tax and transfer schemes, we introduce several key modifications to the baseline DEQN algorithm. At the beginning of each simulation, we initialize the climate-economy state $x_{\text{state},t}$ to a fixed starting point $x_{\text{state},0}$, representing an initial condition such as the year 2015, and simulate forward over a horizon of 350 years. For each simulation, we sample a new set of pseudo-states $\vartheta \in [a, b] \subset \mathbb{R}^{d_\vartheta}$ from the parameter space, which remain constant throughout this particular simulation (cf. Appendix B for more details). This approach enables the neural network to consistently learn policy functions $P(x_{\text{state},t}, \vartheta) \approx \mathcal{N}(x_{\text{state},t}, \vartheta)$, including household savings decisions and value functions. Implementation details, such as the explicit loss function ℓ_ν and the state-transition rules that map x_t to x_{t+1} , are provided in Appendix A.3.

4.2 Welfare Surface Approximation via Gaussian Processes

In this section, we explain how, in Step 2 of our method, we construct computationally efficient GP surrogates for approximating SWFs and, in Step 3, how we determine the tax-and-transfer parameters that maximize the planner’s objective (Equation (36)). We consider two distinct quantities of interest (QoIs): (i) a welfare-improving criterion and (ii) a Pareto-improving criterion.

4.2.1 Social Welfare Objective and Pareto-optimal Parameters

When the goal is to maximize expected welfare, the planner solves

$$\begin{aligned} \vartheta^* = \arg \max_{\vartheta \in [a,b]} \widehat{\text{QoI}}(\vartheta) &= \arg \max_{\vartheta \in [a,b]} \mathbb{E}[W(\vartheta)], \\ \text{s.t. constraints,} \end{aligned} \quad (42)$$

where

$$W(\vartheta) = \sum_{t=-10}^{29} \gamma_t \tilde{U}_t(\vartheta), \quad (43)$$

γ_t are the welfare weights, as defined in Equation (13), and $\tilde{U}_t(\vartheta)$ is a short-hand notation for the expected lifetime utility of generation t as of time 0 (see Equations (10) and (13)), at a particular realization of the pseudo-state variables ϑ .¹³

Recall that the goal of our method’s second step is to find the parameters ϑ that maximize the expected SWF (see, for instance, Equation (42)). A direct approach would involve using a black-box optimizer to evaluate the SWF at numerous candidate parameter vectors, $\{\vartheta_i\}_{i=1}^n$, until a desired numerical accuracy is reached. However, this is computationally prohibitive. Each evaluation of the SWF for a single ϑ_i is expensive, as it requires an estimate derived from 10,000 independent forward simulations over an approximately 150-year horizon. Repeating this costly process for the potentially thousands, if not tens of thousands, of queries required by a constrained optimization routine such as IPOPT (Wächter and Biegler, 2006) would take hours of computation time.

To overcome this computational barrier, we “front-load” the intensive calculations by employing a surrogate modeling approach, as in Step 1. We build a GP surrogate (cf. Section 4.2.2 for an introduction) that approximates the objective function, using only a small number of true SWF evaluations

¹³We assume that the planner’s objective is to maximize the lifetime welfare of all cohorts that are already alive at $t = 0$ as well as those that will be born over the next 150 years; formally, cohorts with birth dates $t \in [-10, 29]$, where $t = -10$ denotes the oldest surviving cohort.

as training points.¹⁴ The predictive mean of this surrogate, $\mu_*(\vartheta)$, then serves as a computationally cheap proxy for the true objective function:

$$\mu_*(\vartheta) \approx \mathbb{E}[W(\vartheta)]. \quad (44)$$

This GP surrogate is then provided to the optimizer. The key advantage lies in the dramatic reduction of computational cost: evaluating the surrogate at a particular ϑ_i has a negligible interpolation cost of $\mathcal{O}(n)$ (Rasmussen and Williams, 2005), where n is the number of training points the GP was fitted to (which in our applications is approximately 450–500 points (cf. Section 5 and Appendices C.2 and C.3)). This is orders of magnitude cheaper than the $\mathcal{O}(\text{paths} \times \text{periods})$ cost of a full Monte Carlo evaluation, which, in our concrete applications, is $10,000 \times 29$ per function evaluation. This efficiency gain of several orders of magnitude enables the rapid optimization in Step 3 (Section 4.2.3) to recover welfare-improving parameters ϑ^* without further costly simulations.

The need for a surrogate becomes even more pronounced when the planner seeks Pareto improvements. In this case, the optimization problem reads as:

$$\vartheta^* = \arg \max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \mathbb{E}_0[\tilde{U}_t(\vartheta)] \quad \text{s.t.} \quad \mathbb{E}_0[\tilde{U}_t(\vartheta)] \geq \mathbb{E}_0[U_t] \quad \forall t, \quad (45)$$

where $\mathbb{E}_0[U_t]$ denotes the lifetime utility of generation t under the BAU scenario, which we compute by solving and simulating a separate version of the model without taxes and transfers. Evaluating the 40 cohort-specific constraints by Monte-Carlo simulation at every candidate ϑ would be computationally prohibitive. To render the problem tractable, we construct a vector-valued surrogate (based on GPs) that approximates the mapping

$$\vartheta \mapsto (\tilde{U}_{-10}(\vartheta), \dots, \tilde{U}_{29}(\vartheta))^\top, \quad (46)$$

that is, one GP per generation. These surrogates deliver instantaneous predictions of both the objective and the constraints, allowing the constrained optimization in Equation (45) to be solved at negligible marginal cost. We next outline how GPR is used to approximate the welfare objective and cohort-specific utilities.

4.2.2 Gaussian Process Regression.

GPR models the target function as a realization of a GP with a mean function $m(\vartheta)$ and a covariance kernel $k(\vartheta, \vartheta')$. Given a training set

$$\mathcal{D} = \{(\vartheta_i, y_i)\}_{i=1}^n = [\Theta, y], \quad \vartheta_i \in [a, b] \subset \mathbb{R}^{d_\vartheta}, \quad y_i \in \mathbb{R}, \quad (47)$$

we aim to learn a mapping $f(\vartheta)$ such that $y_i = f(\vartheta_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ represents independent noise that accounts for simulation variance and ensures numerical stability (Rasmussen and Williams, 2005). Here, ϑ_i are the pseudo-states that parametrize tax-and-transfer schemes, and $y_i = \widehat{\text{QoI}}(\vartheta_i)$ are the welfare outcomes computed via simulations using the neural network surrogate $\mathcal{N}(\cdot)$ for $P(\cdot)$.

¹⁴GPs offer three main advantages in our setting. First, they are highly data-efficient: by encoding smoothness and other prior beliefs through the kernel (for example, a Matérn kernel), a GP can learn nonlinear response surfaces from very few training points. In our own experiments, testing multiple deep neural network architectures, the GP achieved the desired out-of-sample accuracy with approximately ten times fewer model evaluations than the best deep neural net, which still produced slightly less accurate predictions (cf. Appendices C.2 and C.3). Second, GPs provide closed-form posterior variances, yielding well-calibrated uncertainty measures that are invaluable for acquisition rules in BAL, enabling the efficient population of the GP’s training set. Third, the GP’s training, while involving a non-convex optimization to tune kernel hyperparameters, is often more stable and requires tuning far fewer parameters than the stochastic-gradient-based training of a deep neural network. Although deep networks can also be used as surrogates (see, e.g., Chen et al., 2026 and Kase et al., 2022 for economic applications), the GP surrogate is markedly more effective in the high-cost, small-data regime of our welfare evaluations.

In our numerical examples below, we set $m(\vartheta) \equiv 0$ and primarily use the Matérn 5/2 kernel to ensure moderate smoothness and finite differentiability:

$$k(\vartheta, \vartheta') = \sigma_f^2 \left(1 + \frac{\sqrt{5} r}{\rho} + \frac{5r^2}{3\rho^2} \right) \exp\left(-\frac{\sqrt{5} r}{\rho}\right), \quad (48)$$

where $r = \|\vartheta - \vartheta'\|$, σ_f^2 is the signal variance, and ρ is the length-scale.¹⁵ Kernel hyperparameters are optimized by maximizing the marginal likelihood (Murphy, 2022, Ch. 18.2).¹⁶ The posterior distribution at a test input ϑ_* is Gaussian:

$$p(f_* | \vartheta_*, \mathcal{D}) = \mathcal{N}(\mu_*, \sigma_*^2), \quad (49)$$

with predictive mean and variance:

$$\mu_*(\vartheta_*) = k(\vartheta_*, \Theta) [K(\Theta, \Theta) + \sigma_\epsilon^2 I]^{-1} y, \quad (50)$$

$$\sigma_*^2(\vartheta_*) = k(\vartheta_*, \vartheta_*) - k(\vartheta_*, \Theta) [K(\Theta, \Theta) + \sigma_\epsilon^2 I]^{-1} k(\Theta, \vartheta_*), \quad (51)$$

where $K(\Theta, \Theta')$ is the covariance matrix over training inputs. The predictive mean $\mu_*(\vartheta)$ serves as the surrogate $\widehat{\text{QoI}}(\vartheta)$, scaling linearly in n per prediction (Rasmussen and Williams, 2005).

Generating Training Data for the GP. Constructing an accurate GP surrogate requires a training set that covers the admissible parameter space $[a, b] \subset \mathbb{R}^{d_\theta}$ using as few expensive simulations as possible. That is, we aim to achieve a target accuracy with as few training points as possible. We first generate an initial design of about $n = 450$ points with the same uniform sampling scheme employed for DEQN training (see Appendix B). This training set is then refined by adding roughly 50 information-rich points selected through BAL (see Renner and Scheidegger, 2018 and Appendix C.1).

In our numerical experiments (cf. Section 5), generating a single observation $y_i = \widehat{\text{QoI}}(\vartheta_i)$ on one CPU takes about 2 seconds; refitting the GP to the current training set adds a further 20–30 seconds. With a final training set of roughly 500 points, the complete surrogate-construction loop therefore finishes in 30–60 minutes, the upper bound arising when several GP surrogates must be fitted (for example, when we search for Pareto-improving policy vectors). Because each y_i can be produced independently, these wall-clock times fall by one to two orders of magnitude on a modest multi-core or cluster setup.

4.2.3 Surrogate-based Optimization of Welfare Parameters

Once the computationally intensive construction of the SWF surrogate, $\widehat{\text{QoI}}(\vartheta)$, is complete, the final step is to solve the constrained optimization problem (36). This problem is generally non-convex, and thus standard optimization algorithms are not guaranteed to converge to the global optimum. A key advantage of our approach is that, once the GP surrogate model is fitted, any suitable optimization algorithm can be applied to solve Equation (36). This optimization is computationally negligible, typically finishing in under a second. For the specific problem addressed in this paper, the objective function appears to be concave over the feasible region. We confirm this property computationally for the simplest two-dimensional case. For higher-dimensional cases, we launch the optimizer from multiple starting points (see Appendices C.2 and C.3 for more details); all runs converge to the same solution, indicating that the results are robust.

¹⁵For details on hyperparameter selection, see Appendix C.

¹⁶We maximize the marginal log-likelihood to tune the GP’s kernel hyperparameters. Since this is a non-convex optimization problem, we use the Adam optimizer (Kingma and Ba, 2014) with a multi-start strategy to mitigate the risk of converging to a poor local optimum; a common practice in the literature (Murphy, 2022). Specifically, we perform the optimization from 10 randomly chosen starting points. In our applications, we found that the resulting hyperparameters were robust to this procedure, exhibiting little sensitivity to the initial starting point.

Operational summary. For the reader’s convenience, Box 1 collects the key inputs, outputs, and cross-references for each step of the pipeline in Figure 1.

Box 1: Implementation checklist for the three-step pipeline.

1. **Step 1 — DEQN equilibrium surrogate** (Section 4.1; loss function: Eq. (113); hyperparameters: Table 12).
Input: Equilibrium conditions $\mathbf{G}(x, P) = 0$, augmented state $x = [x_{\text{state}}, \vartheta]^\top \in \mathbb{R}^d$ (d up to 33).
Output: Trained neural network $\mathcal{N}(x) \in \mathbb{R}^{22}$ (assets + value functions).
Cost: ~ 4 h on a laptop; minutes on a GPU.
2. **Step 2 — GP welfare surrogate** (Section 4.2; kernel: Eq. (48); active learning: Appendix C.1, Figure 7).
Input: 450–800 welfare evaluations $\{(\vartheta_i, y_i)\}$, each from 10,000 forward simulations of $\mathcal{N}$.
Output: Predictive mean $\mu_*(\vartheta)$ and variance $\sigma_*^2(\vartheta)$ for the SWF (and, if needed, 40 cohort-specific GPs for Pareto constraints).
Cost: 30–60 min (parallelizable).
3. **Step 3 — Constrained policy optimization** (Section 4.2.3; Pareto problem: Eq. (45)).
Input: GP surrogate from Step 2, BAU utilities $\{\mathbb{E}_0[U_t]\}$, linear tax-rate bounds.
Output: Optimal policy parameters ϑ^* .
Cost: < 1 s.

5 Results

In this section, we illustrate how our three-step machine-learning methodology can systematically derive welfare- and Pareto-improving policies within a SOLG IAM. In Section 5.1, we report results for the 17-dimensional SOLG IAM under the BAU scenario, which serves as a baseline for comparison with models incorporating climate policies and transfers. Furthermore, the BAU results provide the necessary benchmark for systematically deriving Pareto-improving tax-and-transfer schemes, as formulated in Equation (45). In Section 5.2, we introduce a simple linear tax on cumulative emissions, accompanied by an exogenously prescribed transfer scheme, which adds two pseudo-state variables; this low-dimensional setting allows a graphical illustration of the method. In Section 5.3, we extend this to include optimal transfers as additional state variables, incorporating 12 transfer parameters alongside the two tax parameters, yielding a state space of $d_x = A + 5 = 17$ and parameter space of $d_\vartheta = 14$, for a total of 31 dimensions. In Section 5.4, we examine a richer tax specification that levies a linear tax on cumulative emissions, carbon intensity, and proximity to the climate tipping point, paired with optimal intergenerational transfers. This setting introduces $d_\vartheta = 16$ pseudo-state variables, for a total of 33 dimensions. Appendix D complements the main results with a targeted sensitivity analysis, exploring how our findings vary under alternative calibrations of the abatement-cost parameter θ_1 and the carbon-intensity process κ_t .

5.1 Solving the Business-As-Usual Model

We compute global solutions to the SOLG IAM under a BAU scenario (see Appendix A.3 for implementation details). The solution algorithm terminates when the loss function falls below a preset threshold, $\ell_\nu^{\text{BAU}} < \epsilon = 10^{-6}$. We then evaluate the model’s out-of-sample accuracy using two standard metrics: the relative Euler equation errors (87) and the relative errors in the value functions (91).

These errors are evaluated over 10,000 individual sample paths, each with a simulation length of 150 years. As reported in Table 2, the mean out-of-sample relative errors are of the order of 10^{-4} , and the maximum errors (defined as the 99.9th percentile) are of the order of 10^{-3} or smaller. This level of accuracy is highly competitive with other global solution methods for stochastic OLG models (see, e.g., Krueger and Kubler, 2006 and Brumm et al., 2017).

We now turn to the model’s solution. Figure 2 presents the main results under the BAU scenario. The top-left panel shows that mean endogenous emissions closely follow the RCP 4.5 trajectory, an

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0003	0.0010	0.0005	0.0004	0.0004	0.0002	0.0013	0.0015	0.0003	0.0004	0.0008
Rel EE 99.9	0.0022	0.0019	0.0024	0.0017	0.0018	0.0013	0.0019	0.0029	0.0016	0.0019	0.0025
Rel Value Function Mean	0.0003	0.0002	0.0003	0.0002	0.0002	0.0002	0.0004	0.0006	0.0002	0.0004	0.0007
Rel Value Function 99.9	0.0025	0.0027	0.0023	0.0025	0.0020	0.0021	0.0022	0.0019	0.0015	0.0020	0.0025

Table 2: Accuracy metrics for the DEQN model solution under the BAU scenario. The table presents the mean and maximum (defined as the 99.9th percentile) for two key error metrics: the relative Euler equation error (Rel EE) and the relative value function error across the various overlapping generations (Gen j). Statistics are computed over 10,000 simulated sample paths.

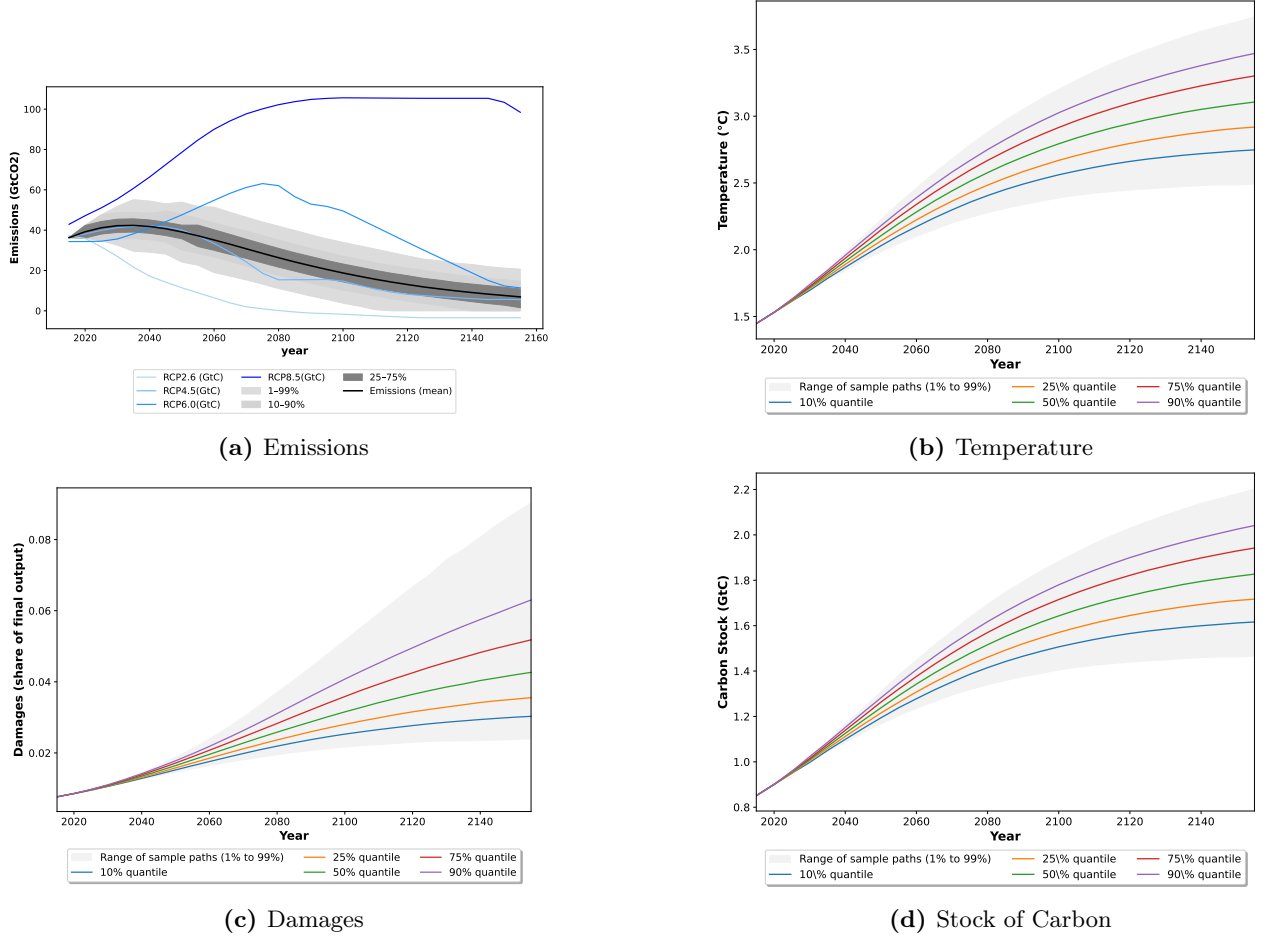


Figure 2: Projected climate and economic outcomes under the BAU scenario. The figure displays key projections from the SOLG IAM over a 150-year horizon. *Top left:* Endogenous emissions (mean and full distribution), with the mean trajectory closely following the RCP 4.5 scenario. *Top right:* Global warming, projected to reach a mean of approximately 3°C, with a substantial uncertainty range of 2.5°C to 3.6°C. *Bottom left:* Economic damages as a percentage of GDP, averaging 4% but with worst-case outcomes exceeding 15%. *Bottom right:* The corresponding stock of carbon in the atmosphere.

intermediate socioeconomic scenario from the Intergovernmental Panel on Climate Change (IPCC).¹⁷ The top-right panel illustrates the resulting evolution of global temperature. While mean global warming is projected to reach approximately 3°C in 150 years, the associated uncertainty is substantial, with the distribution spanning from 2.5°C to over 3.6°C. The economic consequences are depicted in the bottom-left panel, where damages are projected to reach a mean of 4% of GDP over the same period.

¹⁷The Representative Concentration Pathways are standardized scenarios used in climate modeling to project future greenhouse gas concentrations and radiative forcing, ranging from RCP 2.6 (stringent mitigation) to RCP 8.5 (high emissions). See <https://tntcat.iiasa.ac.at/RcpDb> for the related data sets.

In the worst-case scenarios, however, these damages could exceed 15% of GDP. Finally, the bottom-right panel shows the corresponding stock of atmospheric carbon. As the BAU scenario involves no policy optimization, Steps 2 and 3 of our methodology are not applicable.

5.2 Welfare-improving Linear Taxes on Cumulative Emissions

Following the no-policy (BAU) baseline analysis in the previous section, we now illustrate the complete three-step methodology from Section 4 with a simple policy instrument. We assume the planner can implement a carbon tax that is a linear function of cumulative emissions, E_t :

$$\tau_t(E_t) = \vartheta_0 + \vartheta_E E_t. \quad (52)$$

This tax function introduces two parameters, ϑ_0 and ϑ_E , which are treated as pseudo-states and added to the model’s state space. Their permissible ranges $[a, b] \subset \mathbb{R}^2$ are specified in Table 13 in Appendix B.1.1 for the model designated as “Linear in E”.¹⁸ In addition, for illustrative purposes, we assume a specific transfer scheme: the youngest generation receives the largest share of total tax revenue, and the share for each older generation decreases by 10% relative to the next younger one. The shares are normalized to sum to one, which implies a transfer share of 13.94% for the youngest generation. Formally, our transfer scheme, $\mathbb{T}$, is defined as:

$$\mathbb{T}_{t,j} = 0.9^{(j-1)} \cdot 0.1394 \cdot \tau_t e_t, \quad j \in \{1, \dots, 12\}. \quad (53)$$

Running the DEQN algorithm (Step 1 of our methodology) with the loss function from Equation (113) yields highly accurate global solutions, with mean out-of-sample errors for the relative Euler equations (108) and value functions (112) of the order of 10^{-4} and maximum errors of the order of 10^{-3} , as detailed in Table 3.

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0007	0.0004	0.0003	0.0005	0.0005	0.0005	0.0005	0.0012	0.0013	0.0006	0.0006
Rel EE 99.9	0.0031	0.0021	0.0013	0.0027	0.0017	0.0019	0.0018	0.0029	0.0023	0.0019	0.0017
Rel Value Function Mean	0.0004	0.0004	0.0003	0.0003	0.0003	0.0003	0.0004	0.0004	0.0006	0.0006	0.0008
Rel Value Function 99.9	0.0024	0.0023	0.0021	0.0023	0.0026	0.0025	0.0023	0.0019	0.0019	0.0024	0.0023

Table 3: Accuracy metrics for the DEQN model solution under the welfare-improving scenario with a linear tax on cumulative emissions. The table presents the mean and maximum (defined as the 99.9th percentile) for two error metrics: the relative Euler equation error (Rel EE) and the relative value function error across the various overlapping generations (Gen j). Statistics are computed over 10,000 simulated sample paths.

In Step 2 of our method, the neural-network surrogate for the SOLG policies, $\mathcal{N}(\cdot)$, is used to compute the GP surrogate for the SWF. Following the general form of Equation (43), we build a surrogate that maps the tax parameters, $\vartheta = \{\vartheta_0, \vartheta_E\}$, to the SWF. Specifically, we approximate our quantity of interest, $\widehat{\text{QoI}}(\vartheta) = \mathbb{E}[W(\vartheta)]$, by constructing a GP surrogate for which we assume uniform welfare weights ($\gamma_t = 0.025 \forall t$). The resulting predictive mean of the GP, $\mu_*(\vartheta)$, serves as our fast-to-evaluate surrogate for the SWF (cf. Section 4.2.2):

$$\mu_*(\vartheta) = \widehat{\text{QoI}}(\vartheta). \quad (54)$$

To assess its accuracy, we compute the leave-one-out cross-validation error, which yields a value of $3.2 \cdot 10^{-5}$, indicating high accuracy.

Given this accurate surrogate, Step 3 of our method finds the tax parameters that maximize the SWF. To ensure the optimizer does not search outside the surrogate (that is, does not extrapolate),

¹⁸Appendix B.1 provides further details on the derivation of these parameter bounds.

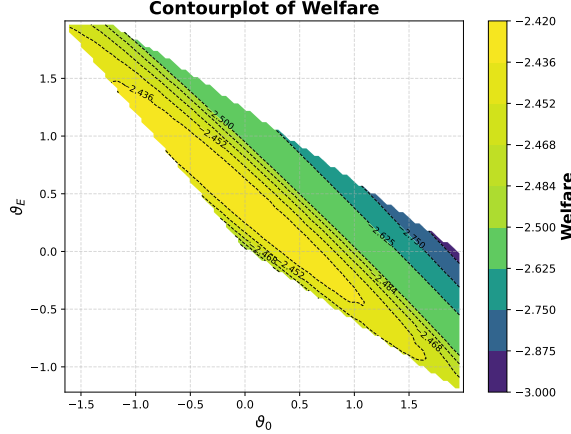


Figure 3: Contour plot of the GP welfare surrogate. The figure displays the predictive mean $\mu_*(\vartheta)$ of the GP surrogate over the two-dimensional tax-parameter space $(\vartheta_0, \vartheta_E)$.

we enforce a set of linear constraints on the tax rate. The resulting constrained optimization problem reads as:

$$\begin{aligned}
 \vartheta^* &= \arg \max_{\vartheta \in [a,b]} \widehat{\text{QoI}}(\vartheta) \\
 &\text{s.t.} \\
 &0 \leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \\
 &0 \leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}.
 \end{aligned} \tag{55}$$

The constraints limit the tax rate in the first and last periods, using cumulative emissions values of $E_0 = 0.851$ and $E_{29} = 1.6$, and a maximum tax rate of $\bar{\tau} = 1.5$ (see Appendix B.1 for details). The resulting surrogate landscape is visualized in Figure 3.

Solving this problem yields the optimal parameters for our tax function (Equation (52)): $\vartheta_0 = -0.01$ and $\vartheta_E = 0.43$. The main results of this welfare-improving tax policy are presented in Figure 4. The top-left panel of the figure shows that the tax policy significantly reduces emissions compared to BAU. For the first 50 years, average emissions are approximately 25% lower than under the BAU scenario; even the 99th percentile of emissions remains below the BAU average. While emissions converge back towards the BAU case after 2100, the policy greatly reduces the risk of extreme emission scenarios. This mitigation effort leads to better climate outcomes, as shown in the middle-left panel. Average global warming stabilizes at approximately 2.7°C, with considerably less uncertainty, ranging between 2.2°C and 3.1°C after 150 years, than in the BAU case.

This reduction in warming translates directly to lower climate damages, as shown in the middle-right panel. Compared to the BAU scenario, damages remain substantially lower, averaging about 3% of GDP and spanning a range of 1.9% to 5% in 150 years. The bottom-right panel shows the corresponding stock of atmospheric carbon, which is maintained at a considerably lower level than in the BAU case. These physical improvements in climate outcomes lead to significant welfare effects across generations, as presented in the top-right panel (cf. Equation (14)). While initial generations (e.g., those entering the workforce in 1985) experience welfare losses of up to 5%, the youngest generations alive at the start of the policy already gain. Welfare gains for future generations increase and stabilize at around 4.8% for those born near 2100. Overall social welfare increases by 1.6% in consumption equivalents compared to the BAU scenario. The policy responsible for these changes is shown in the bottom-left panel. The tax rate starts at approximately 35%, increases over time, and reaches an average of 70% after 150 years, with a range of 55% to 80%.

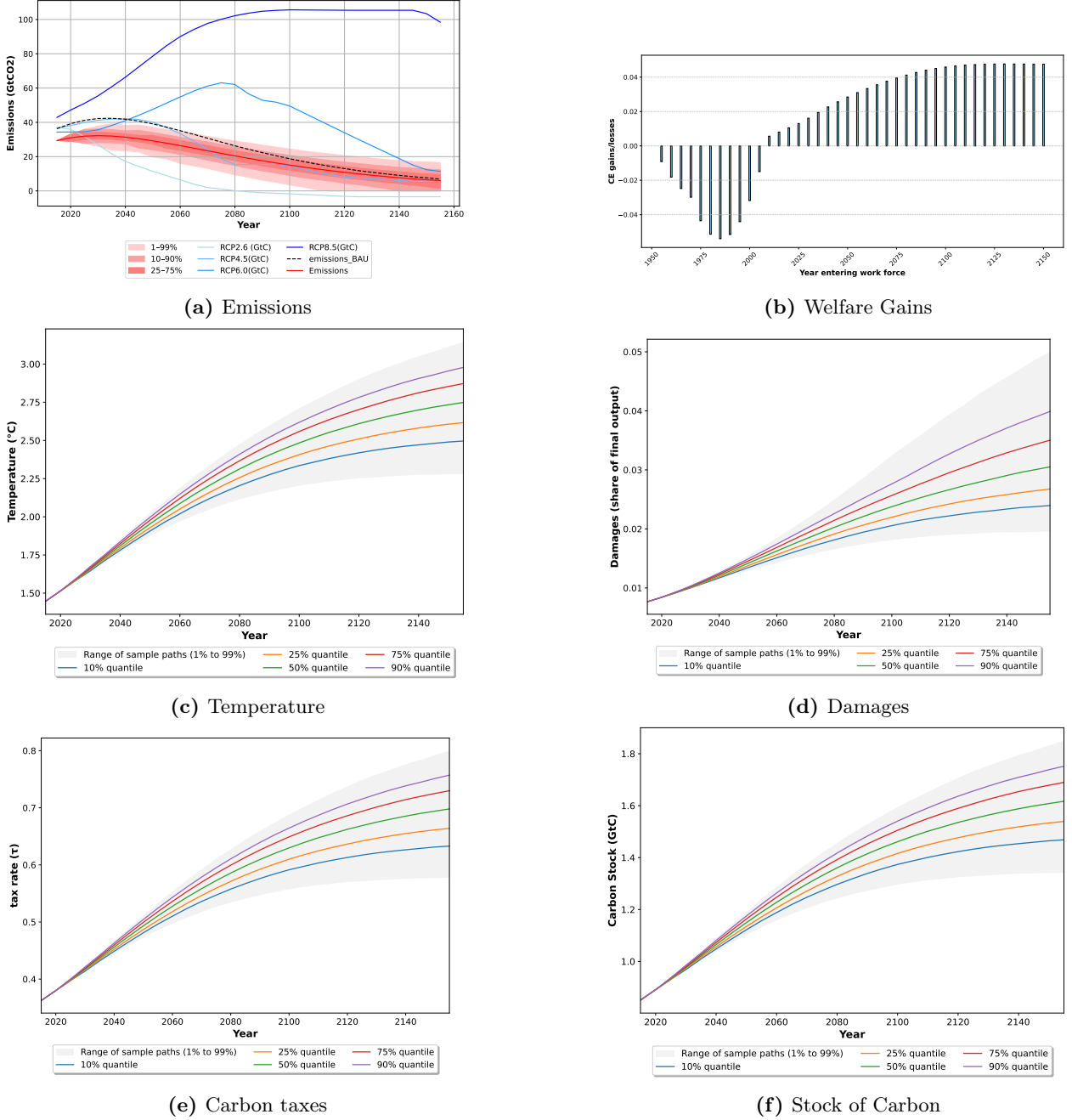


Figure 4: Projected climate and economic outcomes under the optimal tax on cumulative emissions. The figure displays key projections from the SOLG IAM over a 150-year horizon, assuming the optimal tax policy and the exogenous transfer scheme $\mathbb{T}$. *Top left:* Endogenous emissions (mean and full distribution) under the optimal tax policy. *Top right:* Per-generation welfare gains and losses, in consumption equivalents, relative to the no-policy (BAU) scenario. *Middle left:* Global warming, projected to stabilize at a mean of approximately 2.7°C (range: 2.2°C to 3.1°C). *Middle right:* Economic damages as a percentage of GDP, averaging approximately 3%. *Bottom left:* The evolution of the optimal tax rate on cumulative emissions over the 150-year horizon. *Bottom right:* The corresponding evolution of the atmospheric carbon stock.

5.3 Pareto-improving Linear Taxes on Cumulative Emissions and Optimal Transfers

Building on the illustrative example from Section 5.2, we extend the set of policy instruments by allowing the planner to choose transfer shares for each cohort. We apply the procedure from Section 4 to identify Pareto-improving tax-and-transfer policies.

The planner implements a tax that is linear in cumulative emissions, as defined in Equation (52). The sampling of the tax parameters is specified in Table 13 for the model designated as “Linear in E + Transfers.” In addition to the tax, the planner establishes a transfer scheme that allocates a constant share of the resulting tax revenue to each cohort alive in a given period:

$$\mathbb{T}^{\text{planner}} = (\vartheta_1, \vartheta_2, \dots, \vartheta_A)^\top \in \mathbb{R}^A \quad \text{s.t.} \quad \sum_{j=1}^A \vartheta_j = 1, \quad (56)$$

where $A = 12$. The resulting transfer amount for generation j in period t is therefore $\mathbb{T}_{t,j} = \vartheta_j \tau_t e_t$, as per the resource constraint in Equation (12). These transfer shares, ϑ_j , are treated as pseudo-state variables and added to the model’s state space. The shares are sampled from a Dirichlet distribution, as detailed in Appendix B.2. This framework enables a systematic search for Pareto-improving carbon policies.

Applying the DEQN algorithm (Step 1 of our methodology) to this model using the loss function (113) yields highly accurate global solutions. The mean out-of-sample relative Euler equation errors (Equation (108)) and relative value function errors (Equation (112)) are of the order of 10^{-4} , while the maximum errors are of the order of 10^{-3} , as detailed in Table 4.

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0006	0.0002	0.0006	0.0005	0.0001	0.0009	0.0006	0.0003	0.0004	0.0004	0.0003
Rel EE 99.9	0.0035	0.0015	0.0037	0.0043	0.0008	0.0032	0.0014	0.0015	0.0014	0.0013	0.0011
Rel Value Function Mean	0.0003	0.0003	0.0003	0.0003	0.0003	0.0005	0.0003	0.0002	0.0002	0.0004	0.0005
Rel Value Function 99.9	0.0019	0.0018	0.0017	0.0020	0.0019	0.0023	0.0016	0.0013	0.0011	0.0013	0.0018

Table 4: Accuracy metrics for the DEQN model solution in the “Linear in E + Transfers” scenario. The scenario features a linear tax on cumulative emissions and optimal transfers. The table presents the mean and maximum (defined as the 99.9th percentile) for two metrics: the relative Euler equation error (Rel EE) and the relative value function error, evaluated for each overlapping generation (Gen j). Statistics are computed over 10,000 simulated sample paths.

In Step 2, the neural-network surrogate $\mathcal{N}(\cdot)$ provides policy evaluations that are used to train GP surrogates for the SWF. We follow the general setup of Equation (45) and map each cohort’s expected lifetime utility onto the policy-parameter vector

$$\vartheta = \{\vartheta_0, \vartheta_E, \vartheta_1, \dots, \vartheta_{12}\} \in \mathbb{R}^{14}, \quad (57)$$

which contains the two tax coefficients and the twelve transfer shares. Concretely, we fit forty independent GPs, one for each cohort $t = -10, \dots, 29$, according to Equation (46):

$$\mu_{*,t}(\vartheta) = \widehat{\text{QoI}}_t(\vartheta) = \mathbb{E}[\tilde{U}_t^{(2,\mathbb{T})}(\vartheta)], \quad t = -10, \dots, 29. \quad (58)$$

$\tilde{U}_t^{(2,\mathbb{T})}(\vartheta)$ denotes the expected lifetime utility of generation t under the policy regime $(2, \mathbb{T})$, comprising 2 tax instruments and 12 transfers, parametrized by ϑ . Each predictive mean $\mu_{*,t}(\vartheta)$ therefore serves as a computationally inexpensive proxy for the Monte-Carlo estimate of cohort t ’s expected utility under the policy with parameters ϑ .

In Step 3, we solve the constrained optimization problem

$$\begin{aligned}
\vartheta^* &= \arg \max_{\vartheta \in [a, b]} \sum_{t=-10}^{29} \gamma_t \widehat{\text{QoI}}_t(\vartheta) = \arg \max_{\vartheta \in [a, b]} \sum_{t=-10}^{29} \gamma_t \mathbb{E}_0[\tilde{U}_t^{(2, \mathbb{T})}(\vartheta)] \\
\text{s.t.} \quad & \mathbb{E}_0[\tilde{U}_t^{(2, \mathbb{T})}(\vartheta)] \geq \mathbb{E}_0[U_t] \quad \forall t, \\
& 0 \leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \\
& 0 \leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}, \\
& \sum_{j=1}^{12} \vartheta_j = 1,
\end{aligned} \tag{59}$$

where the first set of constraints enforces a Pareto improvement, the next two inequalities bound taxes (as in Equation (55) but with $\bar{\tau} = 0.8$), and the final equality guarantees that the transfer shares sum to one (cf. Appendix B). We set the welfare weights to $\gamma_t = 0.025$ for all t . The optimizer yields the tax coefficients reported in Table 5; the corresponding optimal transfer shares are listed in Table 6.

Coefficient	Symbol	Value
Intercept	ϑ_0	−0.186
Slope	ϑ_E	0.225

Table 5: Pareto-optimal coefficients for the linear tax on cumulative emissions. The table lists the estimated intercept ϑ_0 and slope ϑ_E that jointly maximize social welfare subject to the Pareto constraints.

ϑ_1	ϑ_2	ϑ_3	ϑ_4	ϑ_5	ϑ_6	ϑ_7	ϑ_8	ϑ_9	ϑ_{10}	ϑ_{11}	ϑ_{12}
0.128	0.051	0.058	0.089	0.149	0.09	0.066	0.143	0.076	0.048	0.039	0.061

Table 6: Pareto-optimal transfer scheme under a linear tax on cumulative emissions. The table reports the optimal transfer share allocated to each concurrent cohort j ; shares sum to one.

After solving the SOLG IAM and determining the optimal tax and transfer scheme, we analyze the resulting climate and economic outcomes. Figure 5 presents the main results of this Pareto-improving policy, which combines a linear tax on cumulative emissions with targeted intergenerational transfers.

The top-left panel shows that emissions are, on average, lower than under the BAU scenario, although not by a large margin. The policy’s primary impact is on mitigating upside risk; the 99th percentile of emissions drops below 40 GtCO₂ by 2070, approximately 20 years earlier than in the BAU case. This modest mitigation effort leads to slightly improved climate outcomes, as shown in the middle-left panel. Average global warming stabilizes at a level marginally below 3°C, and importantly, the 99th percentile remains below 3.5°C. The bottom-right panel shows the corresponding stock of atmospheric carbon, which is maintained at a level consistent with these climate outcomes.

This reduction in warming translates to lower climate damages, as illustrated in the middle-right panel. While average damages are only slightly lower than in the BAU scenario, the policy effectively truncates the worst-case outcomes. The 99th percentile of damages reaches approximately 7% of GDP (compared to 9% under BAU), with the worst-case scenarios remaining below 12% (compared to 15% under BAU). Furthermore, the 90th percentile is reduced from 6% to 5%.

These physical outcomes are achieved via a carefully structured tax and transfer scheme designed to ensure Pareto improvement, with welfare effects shown in the top-right panel. The transfers are allocated to shield the initial generations, including those born in the first two periods, from welfare losses, keeping them approximately at their BAU utility levels. Subsequent generations experience progressively larger welfare gains, which increase to nearly 1.4% in consumption equivalents for the last generation considered (cf. Equation (14)). Overall, this policy increases social welfare by 0.42% in

consumption equivalents compared to the BAU scenario. The carbon tax responsible for these changes is depicted in the bottom-left panel.

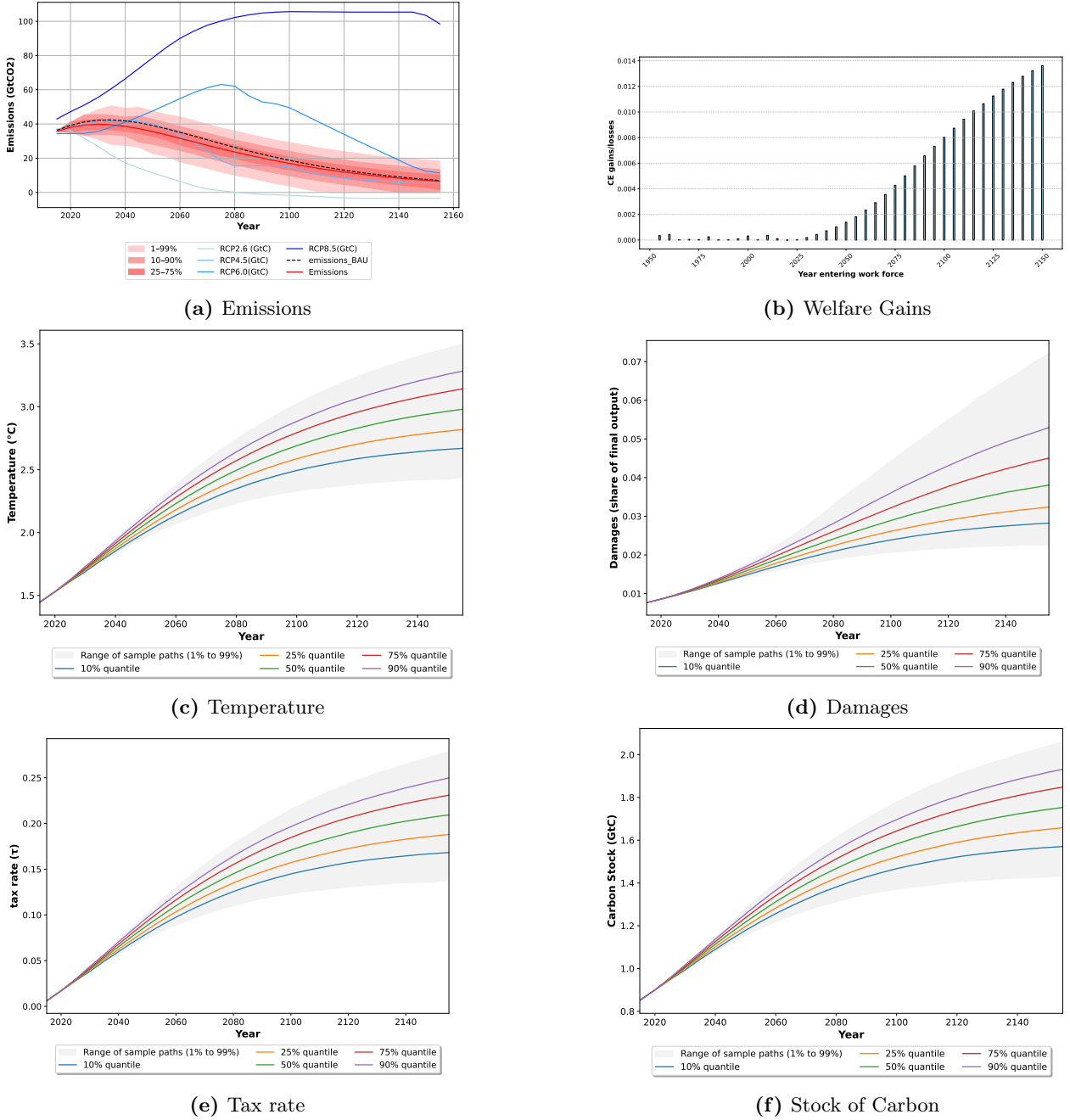


Figure 5: Projected climate and economic outcomes under the optimal tax on cumulative emissions with endogenous transfers. The figure displays key projections from the SOLG IAM over a 150-year horizon for the Pareto-improving tax and transfer policy. *Top left:* Endogenous emissions (mean and full distribution) under the joint optimal tax and transfer scheme. *Top right:* Per-generation welfare gains and losses, in consumption equivalents, relative to the business-as-usual scenario. *Middle left:* Global warming, projected to stabilize at a mean just below 3°C and ranging between 2.3°C to 3.5°C. *Middle right:* Economic damages as a percentage of GDP, averaging below 4% with a 99th percentile of about 7%. *Bottom left:* Evolution of the optimal tax rate on cumulative emissions over the 150-year horizon. *Bottom right:* Corresponding evolution of the atmospheric carbon stock.

5.4 Pareto-improving Linear Taxes on Cumulative Emissions, Carbon Intensity, and Climate Tipping, and Optimal Transfers

In this section, we analyze a markedly richer policy instrument set than in Section 5.3: the planner now controls four levers. The carbon tax is specified as a linear function of cumulative emissions, carbon intensity, and proximity to a climate-tipping point,

$$\tau_t(E_t, \kappa_t, TP_t) = \vartheta_0 + \vartheta_E \frac{E_t}{E_0} + \vartheta_\kappa \frac{\kappa_t}{\kappa_0} + \vartheta_{TP} (1 - \mathbb{D}_{TP}), \quad (60)$$

where the normalized distance to tipping is defined as

$$\mathbb{D}_{TP} = \frac{\max(TP_t - T_t^{AT}, 0)}{TP_{\max} - T_0^{AT}} \in [0, 1], \quad (61)$$

and attains zero once a simulation path crosses the tipping threshold. We set $TP_{\max} = 3.5$ and scale E_t and κ_t by their initial values E_0 and κ_0 .

Equation (60) introduces two additional coefficients, ϑ_κ and ϑ_{TP} , which, together with the other tax and transfer parameters, are treated as pseudo-states and appended to the model’s state vector, raising its dimensionality to $d_\vartheta = 16$. The sampling bounds for all parameters are reported in Table 13 under “Full linear + Transfers,” and the transfer-allocation rule is given in Equation (56).

Applying the same DEQN procedure to the “Full linear + Transfers” specification delivers accuracy that is virtually identical to the simpler model: mean out-of-sample relative Euler equation and value function errors are of the order of 10^{-4} , and the 99.9th-percentile errors are of the order of 10^{-3} (Table 7).

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0004	0.0005	0.0005	0.0003	0.0002	0.0003	0.0004	0.0004	0.0006	0.0006	0.0007
Rel EE 99.9	0.0035	0.0024	0.0032	0.0015	0.0013	0.0030	0.0019	0.0024	0.0022	0.0020	0.0036
Rel Value Function Mean	0.0004	0.0003	0.0002	0.0003	0.0003	0.0004	0.0003	0.0003	0.0004	0.0005	0.0006
Rel Value Function 99.9	0.0016	0.0015	0.0017	0.0019	0.0014	0.0015	0.0013	0.0014	0.0018	0.0020	0.0022

Table 7: Accuracy metrics for the DEQN solution in the “Full linear + Transfers” scenario. The planner’s instruments are a linear tax on cumulative emissions, carbon intensity, and tipping, together with optimal transfers. The table presents the mean and maximum (defined as the 99.9th percentile) for two metrics: the relative Euler equation error (Rel EE) and the relative value function error, for each overlapping generation (Gen j), computed from 10,000 simulated sample paths.

The policy network $\mathcal{N}(\cdot)$ supplies Monte-Carlo estimates of each cohort’s expected lifetime utility, which we interpolate with GPs. The 16-dimensional policy vector now reads as follows:

$$\vartheta = \{\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP}, \vartheta_1, \dots, \vartheta_{12}\} \in \mathbb{R}^{16}, \quad (62)$$

containing four tax coefficients and the twelve transfer shares. As in Equation (59), we fit forty independent GPs, one per cohort $t = -10, \dots, 29$:

$$\mu_{*,t}(\vartheta) = \widehat{\text{QoI}}_t(\vartheta) = \mathbb{E}[\tilde{U}_t^{(4,\mathbb{T})}(\vartheta)], \quad t = -10, \dots, 29. \quad (63)$$

Under the policy regime $(4, \mathbb{T})$, which features four tax instruments and twelve transfers, we denote the expected lifetime utility of generation t by $\tilde{U}_t^{(4,\mathbb{T})}(\vartheta)$. Each predictive mean $\mu_{*,t}(\vartheta)$ serves again as a computationally inexpensive proxy for the corresponding Monte-Carlo utility estimate.

In Step 3, we now solve the Pareto-constrained planner’s problem:

$$\begin{aligned}
\vartheta^* &= \arg \max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \widehat{\text{QoI}}_t(\vartheta) = \arg \max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \mathbb{E}_0[\tilde{U}_t^{(4,\mathbb{T})}(\vartheta)] \\
\text{s.t.} \quad & \mathbb{E}_0[\tilde{U}_t^{(4,\mathbb{T})}(\vartheta)] \geq \mathbb{E}_0[U_t] \quad \forall t, \\
& 0 \leq \vartheta_0 + \vartheta_E \frac{E_0}{E_0} + \vartheta_\kappa \frac{\kappa_0}{\kappa_0} + \vartheta_{TP}(1 - \mathbb{D}_{TP,0}) \leq \bar{\tau}, \\
& 0 \leq \vartheta_0 + \vartheta_E \frac{E_{29}}{E_0} + \vartheta_\kappa \frac{\kappa_{29}}{\kappa_0} + \vartheta_{TP}(1 - \mathbb{D}_{TP,29}) \leq \bar{\tau}, \\
& \sum_{j=1}^{12} \vartheta_j = 1,
\end{aligned} \tag{64}$$

with $E_0 = 0.851$, $E_{29} = 1.6$, $\kappa_0 = 0.35032$, $\kappa_{29} = 0.05$, $\{\mathbb{D}_{TP,0}, \mathbb{D}_{TP,29}\} = \{0.2435, 1.0\}$, and $\bar{\tau} = 0.8$ (cf. Table 13 in Appendix B.1). Uniform welfare weights $\gamma_t = 0.025$ are retained.

The optimizer delivers the tax coefficients reported in Table 8; the optimal transfer shares are listed in Table 9.

Coefficient	Symbol	Value
Intercept	ϑ_0	−0.237
Cumulative emissions	ϑ_E	0.203
Carbon intensity	ϑ_κ	0.037
Distance to tipping	ϑ_{TP}	0.012

Table 8: Pareto-optimal coefficients for the full linear tax on cumulative emissions, carbon intensity, and tipping. The table reports the estimated intercept ϑ_0 and the three slopes $\{\vartheta_E, \vartheta_\kappa, \vartheta_{TP}\}$ that jointly maximize social welfare while satisfying the Pareto constraints.

ϑ_1	ϑ_2	ϑ_3	ϑ_4	ϑ_5	ϑ_6	ϑ_7	ϑ_8	ϑ_9	ϑ_{10}	ϑ_{11}	ϑ_{12}
0.068	0.127	0.064	0.089	0.086	0.156	0.091	0.072	0.065	0.092	0.0	0.088

Table 9: Pareto-optimal transfer scheme with a linear tax on cumulative emissions, carbon intensity, and tipping. The table presents the optimal transfer share allocated to each concurrent cohort j ; shares sum to one (up to rounding).

After obtaining the Pareto-optimal coefficients for the full linear tax on cumulative emissions, carbon intensity, and tipping, along with the associated transfer scheme, we simulate the climate-economy system and analyze the resulting outcomes. Figure 6 summarizes the main results.

The top-left panel shows that mean emissions lie only modestly below the BAU path; the policy’s primary effect is to compress the upper tail. The 99th percentile falls beneath 40 GtCO₂ by 2070, roughly two decades earlier than under BAU. This mitigation translates into slightly improved climate outcomes (*middle-left*): Global warming stabilizes just under 3°C on average, and the 99th percentile remains below 3.5°C.

Lower warming curtails economic damages (*middle-right*). Average losses remain only marginally below BAU, yet extreme outcomes are noticeably truncated: the 99th percentile reaches about 7% of GDP, and the worst observed realizations stay near 11% (versus 9% and 15% in BAU). The 90th percentile declines from roughly 6% to 5%.

Welfare effects appear in the top-right panel. Transfers are calibrated so that cohorts already alive and those born in the first two periods retain utilities close to their BAU levels; several even realize slight gains. Thereafter, welfare rises monotonically, exceeding 1.4% (in consumption-equivalent terms; cf. Equation (14)) for the youngest generation considered. Aggregating across cohorts yields a social-welfare gain of 0.45%, only about 7% larger than the 0.42% gain achieved by the simpler cumulative-

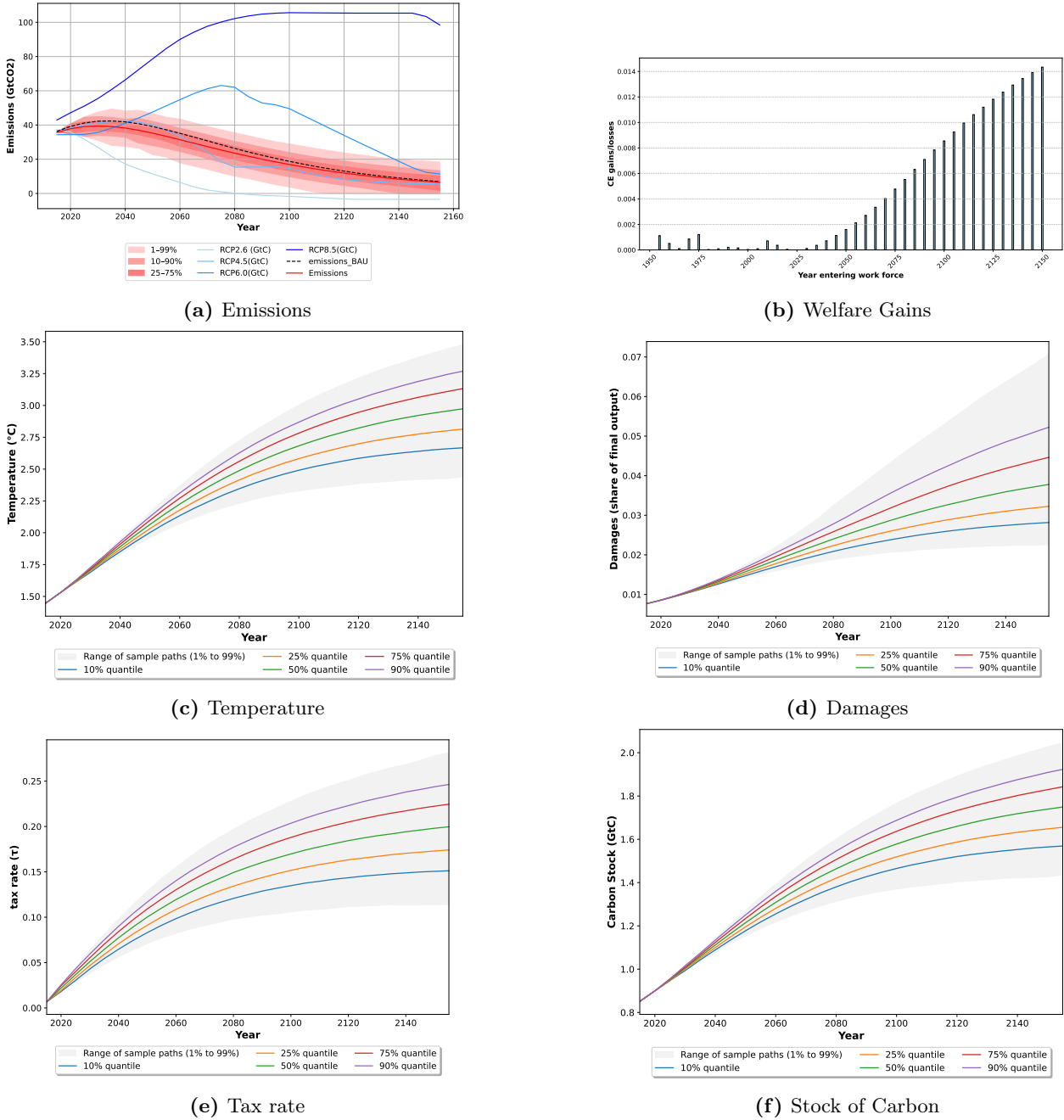


Figure 6: Projected climate and economic outcomes under the optimal linear tax on cumulative emissions, carbon intensity, and tipping with endogenous transfers. The figure reports key projections from the SOLG IAM over a 150-year horizon for the Pareto-improving policy that combines four tax instruments with an optimal intergenerational transfer scheme. *Top left:* Endogenous emissions (mean and full distribution) under the joint optimal tax and transfer scheme. *Top right:* Per-generation welfare gains and losses, in consumption equivalents, relative to the business-as-usual scenario. *Middle left:* Global warming, projected to stabilize at a mean just below 3°C and ranging from 2.3°C to 3.5°C. *Middle right:* Economic damages as a percentage of GDP, averaging below 4% with a 99th percentile of about 7%. *Bottom left:* Evolution of the optimal linear tax rate on cumulative emissions, carbon intensity, and tipping over the 150-year horizon. *Bottom right:* Corresponding evolution of the atmospheric carbon stock.

emissions tax. Hence, augmenting the tax base to include carbon intensity and tipping points provides only marginal additional benefit in this calibration, as documented in Table 10.

	2 instruments	4 instruments
Welfare gain	0.42%	0.45%

Table 10: Aggregate welfare gains under Pareto-improving tax-and-transfer schemes with increasing policy complexity. The table reports the overall consumption-equivalent welfare improvement for two specifications: (i) a policy with two tax instruments, intercept and slope in cumulative emissions (Section 5.3); (ii) a policy with four instruments, taxes linear in cumulative emissions, carbon intensity, and tipping (Section 5.4).

The bottom-left and bottom-right panels plot the optimal tax rate and the implied atmospheric carbon stock, respectively, both of which are consistent with the moderate but effective mitigation profile described above.

6 Conclusion

Designing effective and Pareto-improving climate policy requires navigating complex trade-offs in a world characterized by uncertainty and heterogeneous agents. The computational burden of solving for optimal policies in such settings has traditionally limited quantitative analysis to simplified models. This paper confronts this challenge by developing and applying a scalable three-step computational framework for deriving constrained optimal tax-and-transfer rules in a stochastic overlapping generations model with climate risk. By integrating Deep Equilibrium Nets for global solutions and Gaussian process surrogates for efficient welfare optimization, our methodology makes the systematic search for optimal policies computationally tractable.

Our application of this framework to a stochastic overlapping generations model with uncertain carbon intensity and climate tipping points yields several key insights. First, we confirm that an unconstrained, welfare-maximizing carbon tax, in our case, a simple linear function of cumulative emissions, can substantially reduce climate risk, stabilizing the average temperature increase at 2.7°C compared to 3°C under business-as-usual. However, this policy comes at a significant distributional cost, imposing welfare losses of up to 5% on initial generations, highlighting the political implementation challenge.

The primary contribution of our analysis is demonstrating how to overcome the intergenerational distributional hurdle inherent in optimal climate policy, that is, the fact that welfare-maximizing carbon taxes impose losses on currently living generations whose political support is necessary for implementation. By simultaneously optimizing the tax rule and an accompanying intergenerational transfer scheme, our framework successfully identifies a Pareto-improving policy. This policy guarantees that no generation is made worse off relative to the business-as-usual scenario, shielding initial cohorts from losses through carefully targeted transfers of tax revenue. While the resulting climate mitigation is more modest, this Pareto-improving policy still achieves a notable social welfare gain of 0.42% in consumption-equivalent terms and effectively truncates the worst-case climate and economic damage scenarios. This result underscores that the design of the transfer scheme is as critical as the design of the tax itself for creating implementable climate policy.

Furthermore, our analysis provides valuable insights into the optimal complexity of tax rules. When we expand the tax base from a simple function of cumulative emissions to a richer specification that also includes carbon intensity and proximity to tipping points, we find that the additional welfare gains are surprisingly small. The overall social welfare gain increases from 0.42% to only 0.45% in consumption-equivalent terms. This suggests that a simple, well-designed tax rule, when paired with an optimal transfer system, can capture the vast majority of achievable welfare gains. For policymakers, this suggests focusing on the political economy of implementing simpler, robust rules and their associated transfers, rather than on designing ever more complex tax formulas that may offer little marginal benefit.

While our model is deliberately stylized to isolate key intergenerational mechanisms, the framework we propose is both general and extensible. Future research could incorporate richer features, such as intra-generational heterogeneity, a more detailed multi-sector energy transition, endogenous technological change, or the design of state-dependent transfer schemes to allow for a more ambitious initial tax path. A natural extension would be to explore nonlinear policy rules, such as quadratic forms in the state variables. A quadratic specification would roughly double the number of tax coefficients relative to the linear case, substantially increasing the dimensionality of the pseudo-state space. While this remains computationally feasible within the logic of our framework, it would require considerable additional effort in training the DEQN surrogate. The sharply diminishing returns observed within the linear class, where enriching the tax base from one to three state variables raises the Pareto-improving welfare gain by only 7%, suggest that such refinements may yield limited additional gains, but this remains an open question. It is also an open question which non-linear functions turn out to be optimal in this context.

Computing the first-best (i.e., fully Pareto-efficient) allocation is beyond the scope of this analysis. In highly stylized models (e.g. [Ball and Mankiw, 2007](#); [Gottardi and Kubler, 2011](#)), such allocations can be characterized analytically; it is clear that in our setting, they cannot be achieved through emission taxes alone and require explicit intergenerational transfers.

The methodology developed here provides a template for tackling a wide range of previously intractable Ramsey problems in contemporary macroeconomics, from optimal fiscal and monetary policy in heterogeneous-agent models to the design of social insurance systems. By making the computation of constrained optimal policies in complex environments feasible, this work opens new avenues for quantitative policy analysis.

A Additional Model Details

This appendix first supplies additional modeling details and then explains how the SOLG IAM is mapped onto the DEQN solution framework. Appendix A.1 sets out the recursive formulation of the household problem, Appendix A.2 elaborates on the climate emulator used in the SOLG, and Appendix A.3 describes the transformations that render the model DEQN-amenable.

A.1 Recursive Formulation Of The Household Problem

In order to assess lifetime utility (cf. Section 2.2), we reformulate the household problem into a recursive problem:

$$V_{t,j} = \max_{C_{t,j}, a_{t+1,j+1}} \left\{ \left(\frac{C_{t,j}^{1-\sigma}}{1-\sigma} \right) + \beta \mathbb{E}_t V_{t+1,j+1} \right\} \quad \forall j \in \{1, \dots, A-1\} \quad (65)$$

$$V_{t,A} = \left(\frac{C_{t,A}^{1-\sigma}}{1-\sigma} \right) \quad (66)$$

subject to the budget constraint (9), initial condition $a_{t,1} = 0$, and terminal condition $a_{t+1,A+1} = 0$.

A.2 The Climate Module

Global emissions, e_t , are defined as in (2). The stock of atmospheric carbon, E_t , and the atmospheric temperature anomaly, T_t^{AT} , then evolve according to

$$E_{t+1} = E_t + e_t / \text{C2CO}_2, \quad (67)$$

$$T_t^{AT} = \sigma_{CCR} E_t, \quad (68)$$

where $\text{C2CO}_2 \approx 3.666$ is the conversion factor from carbon to CO_2 and σ_{CCR} is the transient climate response to cumulative carbon emissions (cf. Table 1).

A.3 Deep Equilibrium Nets for SOLG IAMs

In this appendix, we detail the procedure for mapping a stochastic, nonlinear, and non-stationary SOLG IAM presented in Section 2 onto the DEQN framework. Recall that the state of the economy (for the optimal policy case) at time t is given by

$$x_t = [t, TP_t, TP_{\text{reached},t}, \kappa_t, \{a_{t,j}\}_{j=1}^A, E_t, \vartheta]^\top \in \mathbb{R}^{A+5+d_\vartheta}, \quad (69)$$

In the BAU case, the state vector simplifies to

$$x_t^{BAU} = [t, TP_t, TP_{\text{reached},t}, \kappa_t, \{a_{t,j}\}_{j=1}^A, E_t]^\top \in \mathbb{R}^{A+5}, \quad (70)$$

as no tax-and-transfer schemes are present ($\vartheta = \emptyset$) and no mitigation occurs. This reduces the state space dimension by excluding the tax-and-transfer parameters and reflects the absence of policy interventions in the model.

To address the non-stationary nature of the IAM, as in Traeger (2014); Folini et al. (2024), we include time as a state variable and map the unbounded physical time $t \in [0, \infty)$ to a computational time scale via the monotonic transformation

$$t_{\text{comp}} = 1 - \exp(-\zeta t), \quad (71)$$

where $t_{\text{comp}} \in (0, 1]$ and $\zeta > 0$ is a scaling parameter. The physical time is recovered using the inverse

transformation

$$t = -\frac{\ln(1 - t_{\text{comp}})}{\zeta}. \quad (72)$$

In our numerical applications (cf. Section 5), we set $\zeta = 0.015$. This transformation ensures computational tractability while preserving the model's temporal dynamics.

To stabilize the training, we normalize consumption, asset holdings, and capital in effective labor units:

$$c_t = \frac{C_t}{L_t}, \hat{a}_{t,j} = \frac{a_{t,j}}{L_t}, k_t = \frac{K_t}{L_t}, \hat{l}_j = \frac{l_j}{L_t}, v_{t,j} = \frac{V_{t,j}}{L_t^{1/(1-\sigma)}}. \quad (73)$$

Emissions are given by

$$e_t = (1 - \mu_t)\kappa_t L_t k_t^\alpha. \quad (74)$$

The period utility functions (and therefore also the value functions) are scaled with a constant $B = 20$:

$$U(C_t) = U(Bc_t). \quad (75)$$

First-order equations and loss function for the BAU case. Given those normalizations, the first-order equations in the BAU case read as follows:

$$0 = \frac{\left(\beta \mathbb{E}_t \left[(1 + r_{t+1}) c_{t+1,j+1}^{-\sigma} \right]\right)^{-1/\sigma}}{c_{t,j}} - 1, \quad \forall j \in \{1, \dots, A-1\}, \quad (76)$$

$$0 = c_{t,j} + \hat{a}_{t+1,j+1} - (1 + r_t)\hat{a}_{t,j} - w_t \hat{l}_j, \quad \forall j \in \{1, \dots, A\}, \quad (77)$$

$$0 = u_{t,j} + \beta \mathbb{E}_t [v_{t+1,j+1}] - v_{t,j}, \quad \forall j \in \{1, \dots, A-1\}, \quad (78)$$

with the initial conditions $\hat{a}_{t,1} = 0$, and the terminal conditions $\hat{a}_{t+1,A+1} = 0$. In addition, we use the following definitions:

$$k_t = \sum_{j=1}^A \hat{a}_{t,j}, \quad (79)$$

$$1 = \sum_{j=1}^A \hat{l}_j, \quad (80)$$

$$r_t = \Omega_t(T_t^{\text{AT}})\alpha k_t^{\alpha-1} - \delta, \quad (81)$$

$$w_t = \Omega_t(T_t^{\text{AT}})(1 - \alpha)k_t^\alpha, \quad (82)$$

$$c_{t,j} = (1 + r_t)\hat{a}_{t,j} + w_t \hat{l}_j - \hat{a}_{t+1,j+1}, \quad (83)$$

$$u_{t,j} = \frac{c_{t,j}^{1-\sigma}}{1 - \sigma}, \quad (84)$$

$$\sum_{j=1}^A c_{t,j} + \sum_{j=1}^{A-1} \hat{a}_{t+1,j+1} = \Omega_t(T_t^{\text{AT}})(k_t^\alpha) + (1 - \delta)k_t. \quad (85)$$

The (normalized) policy function $P(x_t)$ we intend to approximate with the aid of deep neural networks is given by:

$$\mathcal{N}(x_t^{\text{BAU}}) = [\{\hat{a}_{t+1,j+1}, v_{t,j}\}_{j=1}^{A-1}]^\top \in \mathbb{R}^{2(A-1)}. \quad (86)$$

Having presented the preceding definitions, we can now specify the individual terms that constitute

the loss function used by the DEQN algorithm:

$$l_1, \dots, l_{11} := \frac{\left(\beta \mathbb{E}_t \left[(1 + r_{t+1}) c_{t+1,j+1}^{-\sigma} \right] \right)^{-1/\sigma}}{c_{t,j}} - 1 \quad \forall j \in \{1, \dots, 11\}, \quad (87)$$

$$l_{12} := c_{t,1} + \hat{a}_{t+1,2} - w_t \hat{l}_1, \quad (88)$$

$$l_{13}, \dots, l_{22} := c_{t,j} + \hat{a}_{t+1,j+1} - (1 + r_t) \hat{a}_{t,j} - w_t \hat{l}_j \quad \forall j \in \{2, \dots, 11\}, \quad (89)$$

$$l_{23} := c_{t,12} - (1 + r_t) \hat{a}_{t,12} - w_t \hat{l}_{12}, \quad (90)$$

$$l_{24}, \dots, l_{34} := \frac{u_{t,j} + \beta \mathbb{E}_t [v_{t+1,j+1}]}{v_{t,j}} - 1 \quad \forall j \in \{1, \dots, 11\}. \quad (91)$$

Taken together, these expressions result in the total loss function, that is,

$$\ell_\nu^{BAU} := \frac{1}{N_{\text{path length}}} \sum_{x_t^{BAU} \in \text{sim. path}} \sum_{m=1}^{N_{\text{eq}}=34} \left(\ell_m(x_t^{BAU}, \mathcal{N}(x_t^{BAU})) \right)^2. \quad (92)$$

The final ingredient required by the DEQN algorithm is the one-step state transition for the BAU economy. This transition allows the loss function (92) to be evaluated along simulated trajectories. In the BAU SOLG, the forward mapping for the DEQN simulation is given by

$$x_{t+1}^{BAU} = [t + 1, TP_{t+1}, TP_{\text{reached},t+1}, \kappa_{t+1}, \{a_{t+1,j}\}_{j=1}^A, E_{t+1}]^T. \quad (93)$$

All simulations start in the year 2015 ($t = 0$), so the initial state vector is

$$x_0^{BAU} = [0, 3.0, 0, 0.35032, \{a_{0,j}\}_{j=1}^A, 0.851]^T, \quad (94)$$

corresponding to

$$[t, TP_0, TP_{\text{reached},0}, \kappa_0, \{a_{0,j}\}, E_0]^T. \quad (95)$$

The endogenous state variables are rolled forward according to

$$\hat{a}_{t+1,j} \Rightarrow \hat{a}_{t,j+1}, \quad (96)$$

$$k_{t+1} \Rightarrow k_t, \quad (97)$$

$$E_{t+1} \Rightarrow E_t, \quad (98)$$

$$T_{t+1}^{\text{AT}} \Rightarrow T_t^{\text{AT}}, \quad (99)$$

where the “ $\Rightarrow$ ” indicates that a choice variable at step t becomes an endogenous state at $t+1$. The initial asset distribution is obtained from a deterministic simulation without climate change, and provided in Table 11.¹⁹

$a_{0,1}$	$a_{0,2}$	$a_{0,3}$	$a_{0,4}$	$a_{0,5}$	$a_{0,6}$	$a_{0,7}$	$a_{0,8}$	$a_{0,9}$	$a_{0,10}$	$a_{0,11}$	$a_{0,12}$
0	0.002	0.009	0.021	0.037	0.056	0.076	0.095	0.111	0.093	0.070	0.039

Table 11: Initial asset distribution. Values are obtained from a deterministic simulation without climate change.

First-order equations and loss function for the SOLG with policy. Next, we derive the input necessary for the DEQN for the SOLG model with taxes and transfers. The first-order equations in

¹⁹The initial asset distribution for our simulations is the non-stochastic steady state derived from a simplified, deterministic version of our BAU model in which temperature and damages are held constant at their $t = 0$ levels (see Equation (94)).

this case read as follows:

$$0 = \frac{\left(\beta \mathbb{E}_t \left[(1 + r_{t+1}) c_{t+1,j+1}^{-\sigma} \right]\right)^{-1/\sigma}}{c_{t,j}} - 1, \quad \forall j \in \{1, \dots, A-1\}, \quad (100)$$

$$0 = c_{t,j} + \hat{a}_{t+1,j+1} - (1 + r_t) \hat{a}_{t,j} - w_t \hat{l}_j - \mathbb{T}_{t,j}, \quad \forall j \in \{1, \dots, A\}, \quad (101)$$

$$0 = u_{t,j} + \beta \mathbb{E}_t [v_{t+1,j+1}] - v_{t,j}, \quad \forall j \in \{1, \dots, A-1\}, \quad (102)$$

with initial conditions $\hat{a}_{t,1} = 0$, and terminal conditions $\hat{a}_{t+1,A+1} = 0$. In addition, we use the following definitions:

$$k_t = \sum_{j=1}^A \hat{a}_{t,j}, \quad (103)$$

$$1 = \sum_{j=1}^A \hat{l}_j, \quad (104)$$

$$r_t = \left(\Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t) \right) \alpha k_t^{\alpha-1} - \delta, \quad (105)$$

$$w_t = \left(\Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t) \right) (1 - \alpha) k_t^\alpha, \quad (106)$$

$$\sum_{j=1}^A c_{t,j} + \sum_{j=1}^{A-1} \hat{a}_{t+1,j+1} = \Omega_t(T_t^{AT})(1 - \theta_1 \mu_t^{\theta_2}) k_t^\alpha + (1 - \delta) k_t. \quad (107)$$

As before, the DEQN approximates the normalized policies stated in Equation (86). The individual components entering the loss function for training the DEQN read as follows:

$$l_1, \dots, l_{11} := \frac{\left(\beta \mathbb{E}_t \left[(1 + r_{t+1}) c_{t+1,j+1}^{-\sigma} \right]\right)^{-1/\sigma}}{c_{t,j}} - 1 \quad \forall j \in \{1, \dots, 11\}, \quad (108)$$

$$l_{12} := c_{t,1} + \hat{a}_{t+1,2} - w_t \hat{l}_1 - \mathbb{T}_{t,1}, \quad (109)$$

$$l_{13}, \dots, l_{22} := c_{t,j} + \hat{a}_{t+1,j+1} - (1 + r_t) \hat{a}_{t,j} - w_t \hat{l}_j - \mathbb{T}_{t,j} \quad \forall j \in \{2, \dots, 11\}, \quad (110)$$

$$l_{23} := c_{t,12} - (1 + r_t) \hat{a}_{t,12} - w_t \hat{l}_{12} - \mathbb{T}_{t,12}, \quad (111)$$

$$l_{24}, \dots, l_{34} := \frac{u_{t,j} + \beta \mathbb{E}_t [v_{t+1,j+1}]}{v_{t,j}} - 1 \quad \forall j \in \{1, \dots, 11\}, \quad (112)$$

and result in the total loss function given by

$$\ell_\nu := \frac{1}{N_{\text{path length}}} \sum_{x_t \in \text{sim. path}} \sum_{m=1}^{N_{\text{eq}}=34} (\ell_m(x_t, \mathcal{N}(x_t)))^2, \quad (113)$$

where the state x_t is now given by expression (69), which encompasses the parameters for the tax and transfer schemes. Recall that, in the present SOLG, the state vector is partitioned as $x_t = [x_{\text{state},t}, \vartheta]^\top$. The law of motion for $x_{\text{state},t+1}$ is therefore identical to the BAU specification in Equation (93). The corresponding initial state $x_{\text{state},0}$ is given by Equation (94) together with Table 11.

Updating the pseudo-state vector ϑ within the DEQN framework proceeds as follows. We launch $N = 512$ parallel simulations, all starting from the same climate-economy state $x_{\text{state},0}$ but with distinct draws $\{\vartheta_i\}_{i=1}^N$ from the respective distribution (cf. Section 5). Each simulation is then propagated for $N_{\text{path length}} = 70$ periods under the current tax and transfer scheme. After these trajectories are completed and the network weights have been updated, a new batch of parameter draws is generated, and the process is repeated; ϑ remains fixed along any single trajectory but is resampled before every training episode (cf. Section 4.1).

Hyperparameters. Table 12 summarizes the hyperparameters used in all numerical experiments reported in Section 5. The entry "Parallel simulations" indicates that 512 independent trajectories, each of length 70, are generated per training episode, yielding $512 \times 70 = 35,840$ state transitions before the policies are updated and a new batch of data is produced.

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	10^{-4}
Minibatch size	64
Hidden layers	2
Neurons per hidden layer	512
Activation function	GELU
Path length ($N_{\text{path length}}$)	70
Parallel simulations	512

Table 12: Hyperparameters of the DEQN training routine.

DEQN runtimes. Despite the complexity of the high-dimensional, non-stationary SOLG, DEQN delivers global solutions in remarkably short wall-clock times. On readily available consumer hardware, such as a laptop equipped with an Apple M1 chip, the full training procedure completes in approximately four hours. Running the same code on a high-end accelerator, such as an NVIDIA GH200 Grace Hopper GPU, cuts the runtime to the order of minutes, well below one hour, demonstrating that even modest resources suffice for timely solutions, while specialized hardware can accelerate the process dramatically.

B Domain of the Pseudo-States

Because the SOLG IAM is non-stationary, it is non-trivial to specify a computational domain $[a, b] \subset \mathbb{R}^{d_\vartheta}$ for the pseudo-state vector ϑ over which we build our DEQN and GP surrogate models. The remainder of this appendix details our procedure: Appendix B.1 derives model-consistent bounds for the tax-function parameters, whereas Appendix B.2 establishes the corresponding limits for the optimal transfers.

B.1 Domain of the Tax Parameters

We impose economically meaningful bounds on the tax function, guided by the permissible range of the mitigation effort, $0 \leq \mu_t \leq 1$. First, we enforce a non-negativity constraint on the tax rate ($\tau_t \geq 0$), as negative taxes would effectively subsidize pollution. Second, since mitigation cannot exceed 100% ($\mu_t \leq 1$), any tax rate beyond the level required to achieve full mitigation is economically redundant (see Equation (20)). Together, these two restrictions define the following permissible domain for the tax rate τ_t , which holds for all models:

$$0 \leq \tau_t(\vartheta) \leq \frac{\Omega_t(T_t^{AT})\theta_1\theta_2}{\kappa_t} \quad \forall t. \quad (114)$$

To avoid unnecessary computational costs in the DEQN algorithm (cf. Section A.3), we constrain the parameter space to exclude regions that would produce economically implausible tax policies. Since our framework rules out negative taxes (which would subsidize pollution), parameter combinations that yield negative tax rates are computationally wasteful to explore.

Therefore, we employ a strategy that imposes joint constraints on the tax parameters, ϑ , based on the implied tax rates at the beginning and end of the relevant simulation horizon. Specifically, we enforce bounds on the tax rate, $\tau_t(\vartheta)$, at the initial period ($t = 0$) and at the final period relevant for

welfare evaluations ($t = 29$). This is achieved by evaluating the tax function at heuristic values for the state variables in those two periods, with the state values themselves being informed by simulations of the BAU scenario.

The constraints require the implied tax rate at both $t = 0$ and $t = 29$ to lie within a plausible range $[0, \bar{\tau}]$. This gives rise to the following bounds:

Lower bound. The constraint $\tau_t(\vartheta) \geq 0$ for $t \in \{0, 29\}$ ensures that we discard from the outset any parameter set that implies negative taxes at the start or end of the horizon. The rationale is that a plausible tax path should not begin or end with subsidies for pollution.

Upper bound. Similarly, the constraint $\tau_t(\vartheta) \leq \bar{\tau}$ for $t \in \{0, 29\}$ discards parameter sets that would result in excessively high taxes at these points. The rationale is that an optimal tax path is unlikely to start or end at an extreme level.

Validation. This strategy effectively prunes the parameter space without sacrificing relevant solutions. We validate this approach post hoc, as the optimal tax policies identified in our results (Section 5) are never constrained by these ex ante bounds.

Tax-parameter sampling procedure. We generate candidate parameter vectors for the tax rule in three steps.

(i) *Uniform draw.* Each coefficient is drawn independently from a uniform distribution over the bounds reported in Table 13, which the modeler must specify ex ante. The intervals are deliberately wide so that, during the subsequent optimization, interior solutions are never pushed against these limits; any binding restriction should arise only from the economic constraints (e.g. the requirement that $\tau_0 > 0$), not from the sampling range itself.

(ii) *Feasibility filter.* A draw is retained only if the implied tax rates in the first and last simulated periods lie in economically meaningful intervals,

$$0 \leq \tau_0(\vartheta) \leq \bar{\tau}, \quad 0 \leq \tau_{29}(\vartheta) \leq \bar{\tau}, \quad (115)$$

where the common upper bound $\bar{\tau}$ is set high enough that it never becomes binding in the optimization. To calibrate the bound for the final period, τ_{29} , we evaluate the tax function at representative state values informed by the BAU scenario. For instance, in the model with three tax instruments, these heuristic state values are $E_{29} = 1.6$, $\kappa_{29} = 0.05$, and $\mathbb{D}_{TP,29} = 1$. This check discards parameter draws that would imply economically implausible tax rates, such as negative rates or rates exceeding the level required for full abatement.

(iii) *Objective-specific caps.* The value of $\bar{\tau}$ depends on the planner's objective. When the search is restricted to Pareto-improving policies, the admissible tax range is typically narrower than in exercises that allow some generations to incur welfare losses. The exact bounds used for each model studied in Section 5 are listed in Appendix B.1.1, Table 13.

B.1.1 Heuristic Range of the Tax Parameters

Table 13 provides the permissible ranges for the pseudo-states that parameterize the various tax functions assumed in Section 5.

B.2 Range of Transfer Parameters

The vector of transfer shares, $\mathbb{T}^{\text{planner}} = (\vartheta_1, \dots, \vartheta_{12})^\top$, is sampled from the unit simplex using a Dirichlet distribution, $\mathbb{T}^{\text{planner}} \sim \text{Dir}(\boldsymbol{\alpha})$. This formulation naturally enforces the constraints that all shares are non-negative ($\vartheta_j \geq 0$) and sum to one, that is, $\sum_{j=1}^{12} \vartheta_j = 1$. The vector $\boldsymbol{\alpha} \in \mathbb{R}_{++}^{12}$ contains the concentration parameters that shape the distribution.

A standard choice without prior knowledge would be a uniform distribution over the simplex (i.e., setting $\alpha_j = 1$ for all j). However, this approach is inefficient for our purposes, as it generates highly skewed marginal distributions for each share ($\vartheta_j \sim \text{Beta}(1, 11)$), where large transfers to any single

Model	Planner Target	Parameter	ϑ_i	$\bar{\vartheta}_i$
Linear in E	Max. Welfare	ϑ_0	-2.0	2.0
	Max. Welfare	ϑ_E	-2.0	2.0
	Max. Welfare	$\bar{\tau}$	0.0	1.5
Linear in E + Transfers	Pareto	ϑ_0	-0.5	0.1
	Pareto	ϑ_E	0.0	0.5
	Pareto	$\bar{\tau}$	0.0	0.8
Full linear + Transfers	Pareto	ϑ_0	-1.0	0.5
	Pareto	ϑ_E	-0.5	1.0
	Pareto	ϑ_κ	-0.6	0.6
	Pareto	ϑ_{TP}	-1.0	1.0
	Pareto	$\bar{\tau}$	0.0	0.8

Table 13: Parameter bounds across models and objectives. The columns ϑ_i and $\bar{\vartheta}_i$ denote the lower and upper bounds, respectively. The row labeled $\bar{\tau}$ reports the bounds on the implied tax rate used in the feasibility filter.

generation are rarely sampled. This poor exploration of the policy space’s boundaries can destabilize the DEQN training and hinders the discovery of potentially optimal, highly targeted transfer schemes.

To ensure both stable training and efficient exploration, we adopt a multi-stage sampling strategy. Initially, we use concentration parameters that encourage more uniform draws, preventing any single share from dominating early in the training. For the final analysis, we switch to a more targeted sampling distribution, using the calibrated concentration parameters α shown in Table 14. These values were informed by preliminary optimization runs to focus the search on regions of the policy space likely to yield Pareto improvements.

$\mathbb{T}^{\text{planner}}$	ϑ_1	ϑ_2	ϑ_3	ϑ_4	ϑ_5	ϑ_6	ϑ_7	ϑ_8	ϑ_9	ϑ_{10}	ϑ_{11}	ϑ_{12}
α	1.33	1.25	1.22	0.61	2.7	1.83	0.61	1.79	1.16	0.65	0.67	1.08

Table 14: Dirichlet concentration parameters (α) for transfer share sampling. The parameters govern the sampling distribution used to generate candidate transfer shares for the Pareto-improving policy search.

C Details on Approximating Social Welfare Functions

This appendix describes the construction of the high-accuracy GP surrogates used to identify the SWF-optimal tax and transfer parameters.

Appendix C.1 provides a concise introduction to Bayesian active learning (BAL), a sequential experimental design technique for efficiently selecting training points, and includes an illustrative one-dimensional example of GP regression with active learning. Appendices C.2 and C.3 then document the kernel choices and hyperparameter settings that generate the numerical results reported in Section 5.

C.1 Bayesian Active Learning for Training Point Selection

When training data is expensive to generate, as is the case for us (cf. Section 4.2.1), an efficient sampling strategy is required to build an accurate GP surrogate with a minimal number of training points. While uniform sampling (e.g., via grids or quasi-random sequences) covers the parameter space, it may inefficiently place points in regions where the function is already well-approximated.

BAL offers a principled solution to this problem by sequentially selecting the most informative points for evaluation.

The goal of BAL is to iteratively augment the training dataset, $\mathcal{D}_n = \{(\vartheta_i, y_i)\}_{i=1}^n$, in a way that most rapidly improves the surrogate model and aids in finding the global maximum of the objective function. The BAL process follows an iterative loop:

1. **Train the Surrogate:** Given the current training dataset $\mathcal{D}_n$, we train a GP surrogate. This yields a predictive mean function $\mu_n(\vartheta)$ (our best estimate of the objective function) and a predictive variance function $\sigma_n^2(\vartheta)$ (our uncertainty about that estimate).
2. **Guide the Search:** We use an **acquisition function**, $\alpha(\vartheta)$, to decide which point to evaluate next. This function is cheap to compute and uses the surrogate’s mean and variance to score every potential candidate point ϑ in the parameter space $[a, b] \subset \mathbb{R}^{d_\vartheta}$.
3. **Select the Next Point:** The next point to be evaluated with the expensive simulation, ϑ_{n+1} , is the one that maximizes the acquisition function:

$$\vartheta_{n+1} = \arg \max_{\vartheta \in [a, b]} \alpha(\vartheta). \quad (116)$$

4. **Evaluate and Augment:** We perform one expensive evaluation of the true function to obtain $y_{n+1} = f(\vartheta_{n+1})$. This new pair is added to the training set to create an augmented dataset, $\mathcal{D}_{n+1} = \mathcal{D}_n \cup \{(\vartheta_{n+1}, y_{n+1})\}$.

These steps are repeated until a stopping criterion is met, such as reaching a predefined budget of function evaluations.

Our practical implementation: the upper confidence bound acquisition function. The acquisition function drives the learning process by balancing two opposing objectives. Exploitation steers evaluations toward parameter vectors where the surrogate already predicts a high objective value, sharpening our estimate of the optimum. Exploration, in contrast, targets regions with large predictive uncertainty, thereby reducing overall error and preventing premature convergence to a local optimum. A well-designed acquisition rule blends these forces in a single score, favoring points that are either promising in expected performance or highly informative because of their uncertainty.

A simple and intuitive example that formalizes this trade-off and which we use in our numerical experiments (cf. Section 5) is the Upper Confidence Bound (UCB) acquisition function:

$$\alpha_{\text{UCB}}(\vartheta) = \lambda_{\text{UCB}} \mu_n(\vartheta) + \kappa_{\text{UCB}} \sigma_n(\vartheta). \quad (117)$$

Here, the first term, $\mu_n(\vartheta)$, encourages exploitation, while the second term, $\sigma_n(\vartheta)$ (the predictive standard deviation), encourages exploration. The user-set parameters λ_{UCB} and $\kappa_{\text{UCB}} \geq 0$ control this trade-off. By selecting the point with the highest “optimistic” estimate of the objective, the UCB algorithm efficiently steers the search through the parameter space to find the global maximum. In our practical applications (cf. Section 5), the BAL procedure is implemented as follows. We begin with an initial training set of 450 samples drawn from the parameter space $[a, b] \subset \mathbb{R}^{d_\vartheta}$, on which we train an initial GP. At each iteration of the active learning loop, we set $\lambda_{\text{UCB}} = 1$, $\kappa_{\text{UCB}} = 100$, and evaluate the acquisition function from Equation (117) for a candidate set of 1,000 new points, drawn uniformly from the same domain. The candidate point with the highest acquisition score is then added to the training set, and the GP is re-trained with the augmented data. This iterative process is repeated until the surrogate’s leave-one-out cross-validation error falls below a threshold of 10^{-4} . Further details on the BAL implementation specific to each SOLG IAM are provided in Appendices C.2 and C.3.

Illustrative example: GP regression with active learning in one dimension. To build intuition for the interplay between GP posterior updating and active learning, Figure 7 presents a stylized one-dimensional example. The unknown function (dashed line) is $f(x) = -0.08(x - 1.8)^2 +$

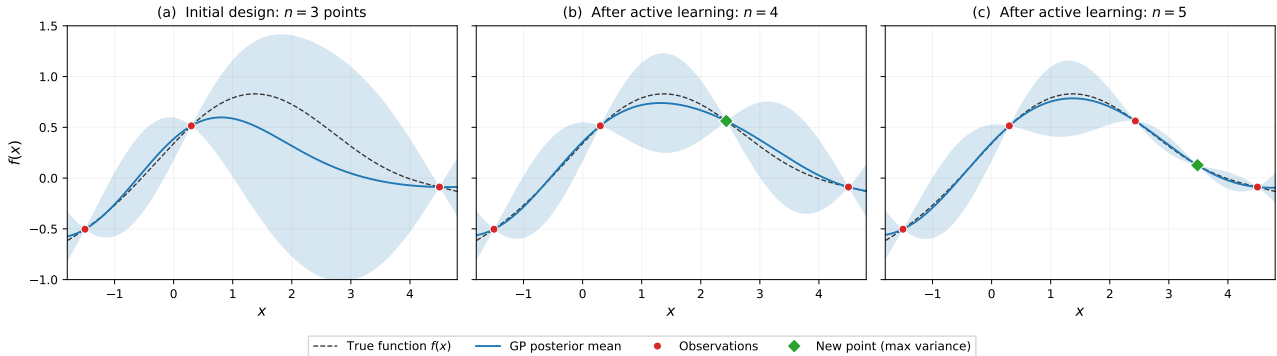


Figure 7: Gaussian process regression with sequential active learning. The unknown function $f(x)$ (dashed) is approximated by a GP (posterior mean: solid blue; 95% credible band: shaded). Red circles denote existing observations; the green diamond marks the point selected by the active-learning criterion (maximum posterior variance). *Left:* Initial design with three evaluations; substantial uncertainty persists in the interior. *Middle:* After one active-learning step ($n = 4$); the credible band narrows markedly in the center. *Right:* After two active-learning steps ($n = 5$); the posterior closely tracks the true function across the entire domain.

$0.25 \sin(1.3x) + 0.6$, a smooth, mildly nonlinear mapping that is qualitatively reminiscent of a welfare objective evaluated over a single policy parameter. We fit a noise-free GP with a squared-exponential kernel (length-scale $\ell = 1.3$, signal variance $\sigma_f^2 = 0.4$) and use the posterior variance as the acquisition criterion, that is, $\alpha(\vartheta) = \sigma_n^2(\vartheta)$, which corresponds to a pure exploration strategy that selects the next evaluation point where the model is most uncertain.²⁰

Left: Initial design with three function evaluations (red circles) at $x \in \{-1.5, 0.3, 4.5\}$. Because the two rightmost points are separated by a wide gap, the GP posterior mean (blue line) deviates substantially from the true function in the interior, and the 95% credible band (shaded region) is correspondingly wide.

Middle: After one round of active learning, the acquisition criterion identifies $x \approx 2.4$ as the point of maximum posterior variance, and the function is evaluated there (green diamond). Incorporating this single observation substantially narrows the credible band in the center of the domain and pulls the posterior mean closer to the true function.

Right: After a second active-learning step, the algorithm selects $x \approx 3.5$, filling the remaining gap. With five evaluations in total, the GP posterior closely tracks the true function across the entire domain, and the residual uncertainty is uniformly small.

This sequential refinement, high uncertainty $\rightarrow$ targeted evaluation $\rightarrow$ rapid posterior contraction, is precisely the mechanism that makes BAL effective in our welfare-surrogate construction. In our numerical experiments in Section 5, each “function evaluation” corresponds to a full Monte-Carlo welfare simulation (10,000 forward paths), and the GP surrogate must interpolate accurately over a d_θ -dimensional policy space (up to 16 dimensions in Section 5.4). BAL ensures that these expensive evaluations are placed where they are most informative, enabling accurate surrogates with training sets of only 450–800 points (cf. Appendices C.2 and C.3).

C.2 Hyperparameters: Welfare-improving Policies

For the welfare-improving surrogate required in Section 5.2, we employ a zero-mean Gaussian process with a Squared Exponential kernel and automatic relevance determination, so that each policy

²⁰In our numerical experiments (cf. Sections 5.2–5.4), we employ, as discussed above, the UCB acquisition function (117), which combines exploration (posterior variance) with exploitation (posterior mean). For the purpose of this illustration, a variance-only criterion isolates the uncertainty-reduction mechanism most clearly.

parameter is endowed with its own length scale. The initial training design comprises 450 parameter vectors drawn uniformly from the admissible hypercube (cf. Appendix B.1.1). Bayesian active learning with $\lambda_{\text{UCB}} = 1$ and $\kappa_{\text{UCB}} = 100$ then adds a further 50 information-rich points, resulting in a final training set of 500 observations.

Kernel hyperparameters are estimated by maximizing the log marginal likelihood via a sequential quadratic programming algorithm. The resulting leave-one-out cross-validation error is typically of the order of 10^{-4} ; for the baseline model discussed in Section 5.2, it falls to 3×10^{-5} .

For comparison, we also experimented with feed-forward neural network surrogates. Across a sweep of ten architectures (best-performing specification: two hidden layers with 128 GELU units, Adam optimizer, learning rate 10^{-4}), Gaussian process regression achieved comparable out-of-sample accuracy with roughly an order of magnitude fewer model evaluations, while delivering slightly lower prediction errors.

C.3 Hyperparameters: Pareto-improving Policies

In the models featuring Pareto-improving tax and transfer schemes, we fit forty independent Gaussian process surrogates, one for each cohort, to approximate the social welfare function. As in the welfare-improving case, we employ a Squared Exponential kernel with automatic relevance determination and the same Bayesian active learning configuration described in Section C.2. Finally, in our sensitivity analysis in Section D we employ a Matérn-5/2 kernel and automatic relevance detection.

Because these models are more complex (cf. Sections 5.3 and 5.4), larger training sets are required to attain comparable accuracy. Final sample sizes range from 500 to 800 observations, depending on the dimensionality of the policy space, until the desired leave-one-out error threshold is reached.

Kernel hyperparameters are estimated using the same likelihood-based optimization procedure as in the welfare-improving case.

D Additional Results

In this appendix, we examine the sensitivity of our main results (cf. Section 5) to alternative values of the abatement-cost parameter θ_1 (Section D.1) and to the carbon-intensity process κ_t (Section D.2).

D.1 Sensitivity to the Abatement Cost Parameter θ_1

We re-solve the Pareto-improving linear-tax-on-cumulative-emissions model of Section 5.3 after lowering the abatement-cost intercept from $\theta_1 = 0.7$ (baseline calibration) to $\theta_1 = 0.6$. All other calibration parameters, the tax functional form, the transfer structure, and the three-step computational pipeline remain unchanged. We therefore focus the discussion exclusively on the differences induced by lower mitigation costs.

Applying the DEQN algorithm (Step 1) to this specification again delivers highly accurate global solutions. Mean out-of-sample relative Euler equation and value function errors are of the order of 10^{-4} , while the 99.9th-percentile errors remain of the order of 10^{-3} (Table 15), indicating that solution quality is maintained under this alternative calibration.

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0005	0.0004	0.0005	0.0005	0.0011	0.0010	0.0006	0.0004	0.0009	0.0005	0.0006
Rel EE 99.9	0.0022	0.0021	0.0026	0.0019	0.0057	0.0052	0.0028	0.0012	0.0021	0.0014	0.0021
Rel Value Function Mean	0.0004	0.0003	0.0004	0.0002	0.0003	0.0005	0.0002	0.0002	0.0007	0.0004	0.0004
Rel Value Function 99.9	0.0015	0.0014	0.0017	0.0013	0.0013	0.0027	0.0014	0.0011	0.0016	0.0013	0.0019

Table 15: Accuracy metrics for the DEQN model solution in a Pareto-improving scenario with $\theta_1 = 0.6$. The table reports the mean and maximum (99.9th percentile) relative Euler equation and value function errors across overlapping generations, computed from 10,000 simulated sample paths.

The Pareto-optimal tax coefficients are $\vartheta_0 = -0.262$ and $\vartheta_E = 0.313$ (Table 16), with the corresponding transfer shares reported in Table 17. Relative to the baseline ($\vartheta_0 = -0.186$, $\vartheta_E = 0.225$), the slope on cumulative emissions is substantially steeper and the intercept more negative. As in Section 5.3, the initial tax rate remains close to zero, reflecting the Pareto constraint for currently alive cohorts. Over time, however, the tax rises more steeply and reaches an average level of approximately 0.27 by 2150, compared with about 0.20 under the baseline calibration. This adjustment is economically intuitive: when abatement is cheaper, the planner can implement a more ambitious emissions-reduction path without violating the Pareto constraints.

Coefficient	Symbol	Value
Intercept	ϑ_0	-0.262
Slope	ϑ_E	0.313

Table 16: Pareto-optimal coefficients for the linear tax on cumulative emissions with $\theta_1 = 0.6$. The table lists the estimated intercept ϑ_0 and slope ϑ_E that jointly maximize social welfare subject to the Pareto constraints.

ϑ_1	ϑ_2	ϑ_3	ϑ_4	ϑ_5	ϑ_6	ϑ_7	ϑ_8	ϑ_9	ϑ_{10}	ϑ_{11}	ϑ_{12}
0.116	0.055	0.144	0.000	0.064	0.246	0.000	0.137	0.088	0.036	0.057	0.053

Table 17: Pareto-optimal transfer scheme under a linear tax on cumulative emissions with $\theta_1 = 0.6$. The table reports the optimal transfer share allocated to each concurrent cohort j ; shares sum to one (up to rounding).

After determining the optimal tax and transfer coefficients, we simulate the climate-economy system and analyze the resulting outcomes. Figure 8 summarizes the main results.

Relative to the baseline Pareto-improving policy, the combination of higher taxes and lower mitigation costs yields modestly lower emissions (top-left panel). Welfare effects are shown in the top-right panel. As in Section 5.3, the optimal transfer scheme shields initial cohorts, keeping their welfare gains close to zero, while several early generations realize small positive gains. Future generations benefit substantially more: the consumption-equivalent welfare gain for the youngest cohort considered reaches approximately 1.75%, compared with about 1.4% under the baseline (cf. Equation (14)). Aggregating across cohorts, social welfare increases by 0.54% in consumption-equivalent terms relative to BAU, compared with a 0.42% gain under the baseline calibration.

Climate outcomes improve accordingly. Mean global warming stabilizes at approximately 2.93°C, with the 99th percentile at 3.4°C, compared with just below 3°C and 3.5°C, respectively, under the baseline. These lower temperatures reduce average economic damages to about 3.8% of final output, down from roughly 4% in Section 5.3. The bottom-left panel shows the evolution of the optimal tax rate, while the bottom-right panel reports the corresponding stock of atmospheric carbon, both consistent with the more ambitious mitigation profile enabled by cheaper abatement.

Overall, these results confirm the expected qualitative response to lower abatement costs while providing a quantitative measure of the model’s sensitivity. A reduction of approximately 14% in θ_1 (from 0.7 to 0.6) raises aggregate welfare gains by roughly 29% (from 0.42% to 0.54%). This nonlinear amplification reflects the interaction of lower abatement costs with convex climate damages and tipping dynamics: cheaper abatement permits a steeper Pareto-feasible tax path, which disproportionately reduces exposure to high-damage realizations associated with the climate tipping region, even though changes in mean emissions and temperatures are moderate. Importantly, while the magnitude of welfare gains varies with θ_1 , the qualitative structure of the Pareto-improving policy, low initial taxes, rising over time, and transfers that protect early cohorts, remains unchanged.

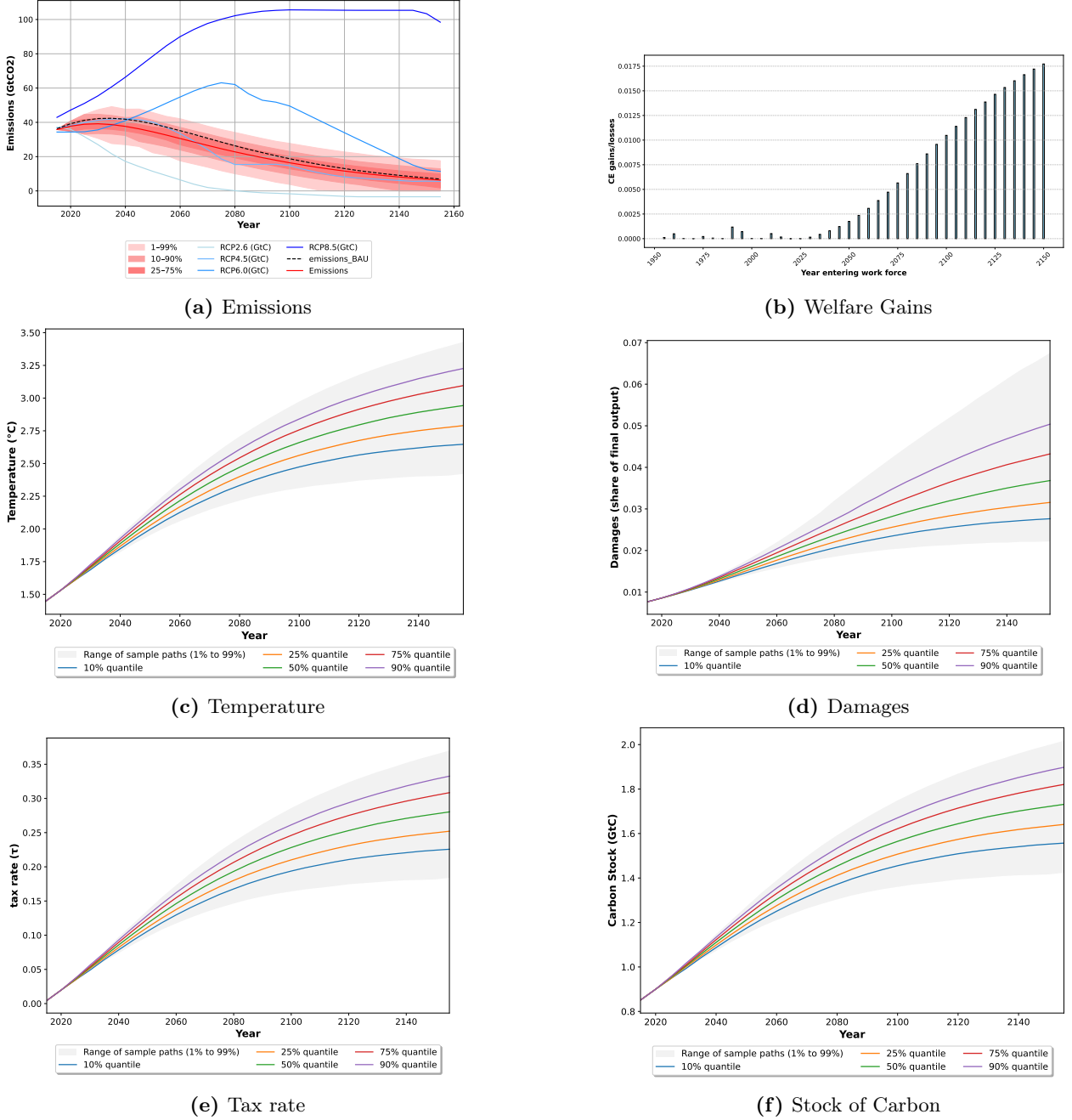


Figure 8: Projected climate and economic outcomes under the optimal tax on cumulative emissions with endogenous transfers ($\theta_1 = 0.6$). The figure displays key projections from the SOLG IAM over a 150-year horizon for the Pareto-improving tax and transfer policy. *Top left:* Endogenous emissions (mean and full distribution) under the joint optimal tax and transfer scheme. *Top right:* Per-generation welfare gains and losses, in consumption equivalents, relative to the business-as-usual scenario. *Middle left:* Global warming, projected to stabilize at a mean of about 2.9°C (range: 2.3°C to 3.4°C). *Middle right:* Economic damages as a percentage of GDP, averaging 3.8% with a 99th percentile of below 7%. *Bottom left:* Evolution of the optimal tax rate on cumulative emissions over the 150-year horizon. *Bottom right:* Corresponding evolution of the atmospheric carbon stock.

D.2 Sensitivity to the Carbon-Intensity Process κ_t

We next examine the sensitivity of our results to the volatility of carbon intensity, maintaining the remaining calibration of the Pareto-improving model in Section 5.3. In the baseline calibration (cf. Section 3), the innovation $\epsilon_{\kappa,t}$ takes the values $\{-0.03, 0, 0.03\}$, each with probability $\frac{1}{3}$. We now

widen the support by approximately 16.6%, setting

$$\epsilon_{\kappa,t} = \begin{cases} 0.035, & \text{with probability } \frac{1}{3}, \\ 0.00, & \text{with probability } \frac{1}{3}, \\ -0.035, & \text{with probability } \frac{1}{3}. \end{cases} \quad (118)$$

Because the distribution of shocks enters agents' expectations, a change in the shock support alters both lifetime utilities and optimal savings decisions. A new BAU solution is therefore required to obtain the correct benchmark utilities $\{\mathbb{E}_0[U_t]\}$ against which Pareto improvements are evaluated, as well as updated policy functions that reflect the higher uncertainty faced by all cohorts. All other calibration parameters, the tax functional form, the transfer structure, and the three-step computational pipeline remain unchanged.

D.2.1 Business-as-Usual under Wider Carbon-Intensity Shocks

Figure 9 displays selected BAU outcomes under the wider shock support. The increased volatility of κ_t generates a noticeably wider fan of emission and temperature paths relative to the baseline BAU (cf. Figure 2). While mean trajectories are virtually unchanged, the tails diverge: the 1st and 99th percentiles of global warming reach approximately 2.4°C and 3.8°C by the end of the 150-year horizon, compared with 2.5°C and 3.6°C under the baseline calibration. The wider temperature distribution implies a greater probability of entering the climate-tipping region, amplifying the role of tail risk in the welfare evaluation. Higher volatility in carbon intensity leads to a wider bandwidth of emissions compared to the baseline calibration. The 1st percentile reaches zero approximately 10 years earlier in comparison to the baseline scenario. However, the 99th percentile is still at 25 $GtCO_2$ after 150 years, compared to around 20 $GtCO_2$ in the baseline calibration.

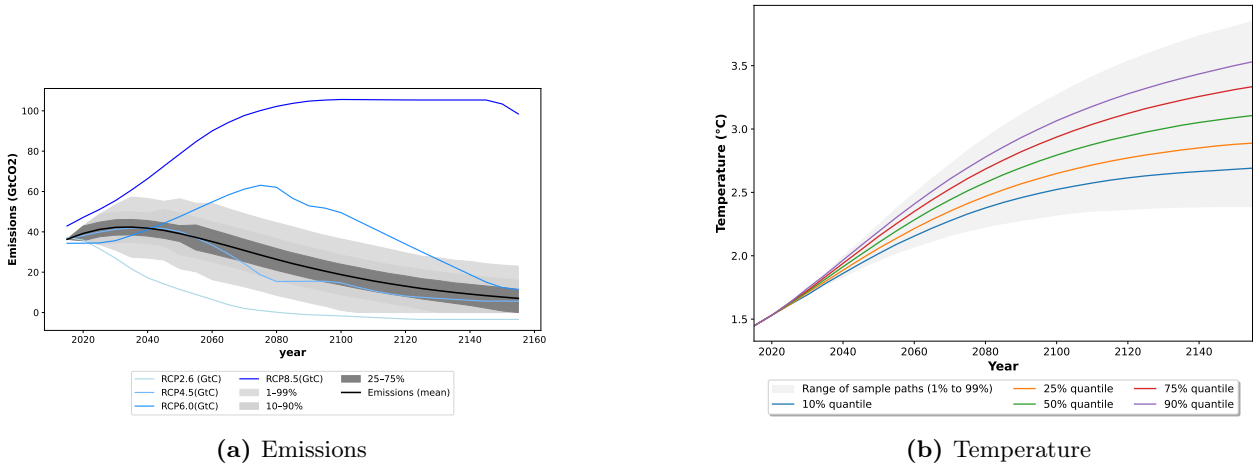


Figure 9: Projected climate outcomes under the BAU scenario with higher carbon-intensity volatility ($\epsilon_{\kappa} \in \{-0.035, 0, 0.035\}$). *Left:* Endogenous emissions (mean and full distribution). *Right:* Global warming, with the 1st–99th percentile range widening to approximately 2.4°C to 3.8°C, compared with 2.5°C to 3.6°C under the baseline calibration.

D.2.2 Pareto-Improving Policy under Wider Carbon-Intensity Shocks

We re-solve the Pareto-improving linear-tax-on-cumulative-emissions model of Section 5.3 under the wider shock support. Applying the DEQN algorithm (Step 1) again delivers highly accurate global solutions: mean out-of-sample relative Euler equation and value function errors are of the order of 10^{-4} , while the 99.9th-percentile errors remain of the order of 10^{-3} (Table 18).

	Gen 1	Gen 2	Gen 3	Gen 4	Gen 5	Gen 6	Gen 7	Gen 8	Gen 9	Gen 10	Gen 11
Rel EE Mean	0.0005	0.0004	0.0005	0.0005	0.0007	0.0008	0.0008	0.0003	0.0010	0.0005	0.0021
Rel EE 99.9	0.0025	0.0026	0.0015	0.0041	0.0061	0.0034	0.0046	0.0021	0.0032	0.0029	0.0104
Rel Value Function Mean	0.0004	0.0004	0.0004	0.0002	0.0003	0.0005	0.0003	0.0002	0.0003	0.0005	0.0008
Rel Value Function 99.9	0.0021	0.0022	0.0020	0.0019	0.0020	0.0026	0.0016	0.0018	0.0013	0.0034	0.0033

Table 18: Accuracy metrics for the DEQN model solution in a Pareto-improving scenario with $\epsilon_\kappa \in \{-0.035, 0, 0.035\}$. The table reports the mean and maximum (99.9th percentile) relative Euler equation and value function errors across overlapping generations, computed from 10,000 simulated sample paths.

The Pareto-optimal tax coefficients are $\vartheta_0 = -0.184$ and $\vartheta_E = 0.220$ (Table 19), with the corresponding transfer shares reported in Table 20. These values are strikingly close to their baseline counterparts ($\vartheta_0 = -0.186$, $\vartheta_E = 0.225$), indicating that the optimal tax rule is largely insensitive to the degree of carbon-intensity volatility within the range considered. This finding is intuitive: the tax is levied on cumulative emissions E_t , which aggregates over many periods and thereby smooths out the additional period-by-period volatility in κ_t .

Coefficient	Symbol	Value
Intercept	ϑ_0	-0.184
Slope	ϑ_E	0.220

Table 19: Pareto-optimal coefficients for the linear tax on cumulative emissions with $\epsilon_\kappa \in \{-0.035, 0, 0.035\}$. The table lists the estimated intercept ϑ_0 and slope ϑ_E that jointly maximize social welfare subject to the Pareto constraints.

ϑ_1	ϑ_2	ϑ_3	ϑ_4	ϑ_5	ϑ_6	ϑ_7	ϑ_8	ϑ_9	ϑ_{10}	ϑ_{11}	ϑ_{12}
0.113	0.076	0.098	0.033	0.116	0.169	0.000	0.157	0.099	0.006	0.000	0.133

Table 20: Pareto-optimal transfer scheme under a linear tax on cumulative emissions with $\epsilon_\kappa \in \{-0.035, 0, 0.035\}$. The table reports the optimal transfer share allocated to each concurrent cohort j ; shares sum to one (up to rounding).

Figure 10 displays the emission and temperature outcomes under the Pareto-improving policy. Mean emissions lie close to the baseline policy path, but the distribution is wider, mirroring the greater underlying volatility. Global mean warming stabilizes near 3°C, though the wider tails of the temperature distribution, reflecting the higher carbon-intensity volatility, are partially compressed by the tax policy.

The welfare and economic outcomes mirror the baseline pattern: the transfer scheme shields initial cohorts while future generations enjoy progressively larger gains. The consumption-equivalent welfare gain for the youngest cohort considered reaches approximately 1.4%. Aggregating across cohorts, social welfare increases by 0.43% relative to BAU (cf. Equation (14)), which is marginally higher than the 0.42% obtained under the baseline calibration in Section 5.3. The marginally higher gain reflects the greater tail risk under BAU when κ_t is more volatile, which the tax policy partially compresses. Economic damages average 4% of GDP, with the 99th percentile at approximately 7.7%. The evolution of the optimal tax rate and the atmospheric carbon stock are both consistent with the moderate mitigation profile observed under the baseline calibration.

Overall, the near-identical tax coefficients and qualitatively unchanged policy structure confirm that our main findings are not sensitive to the degree of carbon-intensity volatility within the range examined. The Pareto-improving policy, low initial taxes rising over time, combined with targeted intergenerational transfers, remains the preferred instrument mix. The principal effect of higher κ_t volatility is to widen the distribution of climate and economic outcomes, but the optimal policy response, as reflected in the tax coefficients, is largely unchanged because the tax is indexed to cumulative

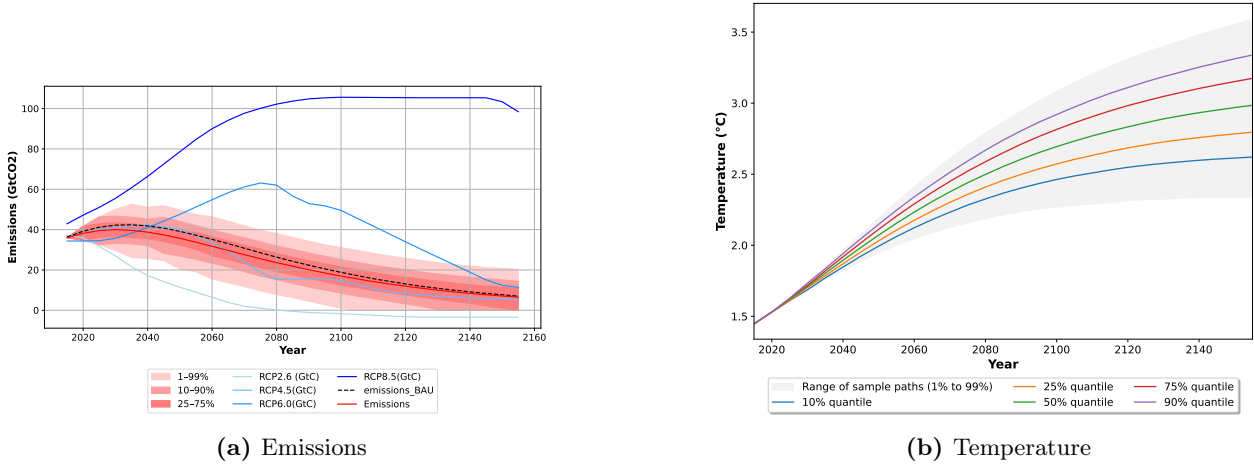


Figure 10: Projected climate outcomes under the optimal tax on cumulative emissions with endogenous transfers ($\epsilon_\kappa \in \{-0.035, 0, 0.035\}$). The figure displays selected projections from the SOLG IAM over a 150-year horizon for the Pareto-improving tax and transfer policy under higher carbon-intensity volatility. *Left:* Endogenous emissions (mean and full distribution) under the joint optimal tax and transfer scheme. *Right:* Global warming, projected to stabilize at a mean near 3°C (range: 2.3°C to 3.6°C).

emissions, which are only modestly affected by the additional period-level noise.

E List of Acronyms

Acronym	Description
BAL	Bayesian active learning
BAU	Business-as-usual
CCR	Carbon-climate response
CRRA	Constant relative risk aversion
CO ₂	Carbon dioxide
DEQN	Deep Equilibrium Nets
FNN	Feed-forward neural network
GDP	Gross domestic product
GP	Gaussian process
GPR	Gaussian process regression
IAM	Integrated assessment model
IPCC	Intergovernmental Panel on Climate Change
OLG	Overlapping generations
QoI	Quantity of interest
RCP	Representative concentration pathway
SOLG	Stochastic overlapping generations
SWF	Social welfare function
TCRE	Transient climate response to cumulative carbon emissions
UCB	Upper confidence bound

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